

Service Manual

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SAFETY CONSIDERATIONS

Installing, starting up, and servicing air-conditioning equipment can be hazardous due to system pressures, electrical components, and equipment location (roofs, elevated structures, etc.). Only trained, qualified installers and service mechanics should install, start-up, and service this equipment.

Untrained personnel can perform basic maintenance functions such as cleaning coils. All other operations should be performed by trained service personnel. When working on the equipment, observe the precautions in the literature and on tags, stickers, and labels attached to the equipment.

Follow all safety codes. Wear safety glasses and work gloves. Keep a quenching cloth and fire extinguisher nearby when brazing. Use care in handling, rigging, and setting bulky equipment.

Read this manual thoroughly and follow all the warnings or cautions included in literature and attached to the unit. Consult local building codes and National Electrical Code (NEC) for special requirements.

Recognize safety information. This is the safety-alert symbol . When you see this symbol on the unit and in instructions or manuals, be alert to the potential for personal injury. Understand these signal words: **DANGER**, **WARNING**, and **CAUTION**.

These words are used with the safety-alert symbol. **DANGER** identifies the most serious hazards which will result in severe personal injury or death. **WARNING** signifies hazards which could result in personal injury or death. **CAUTION** is used to identify unsafe practices which may result in minor personal injury or product and property damage. **NOTE** is used to highlight suggestions which will result in enhanced installation, reliability, or operation.



WARNING

ELECTRICAL SHOCK HAZARD

Failure to follow this warning could result in personal injury or death.

Before installing, modifying, or servicing the system, the main electrical disconnect switch must be in the **OFF** position. There may be more than 1 disconnect switch. Lock out and tag switch with a suitable warning label.



WARNING



EXPLOSION HAZARD

Failure to follow this warning could result in death, serious personal injury, and/or property damage.

Never use air or gases containing oxygen for leak testing or operating refrigerant compressors. Pressurized mixtures of air or gases containing oxygen can lead to an explosion.



CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Do not bury more than 36 in. (914 mm) of refrigerant pipe in the ground. If any section of pipe is buried, there must be a 6 in. (152 mm) vertical rise to the valve connections on the outdoor units. If more than the recommended length is buried, refrigerant may migrate to the cooler buried section during extended periods of system shutdown. This causes refrigerant slugging and could possibly damage the compressor at start-up.

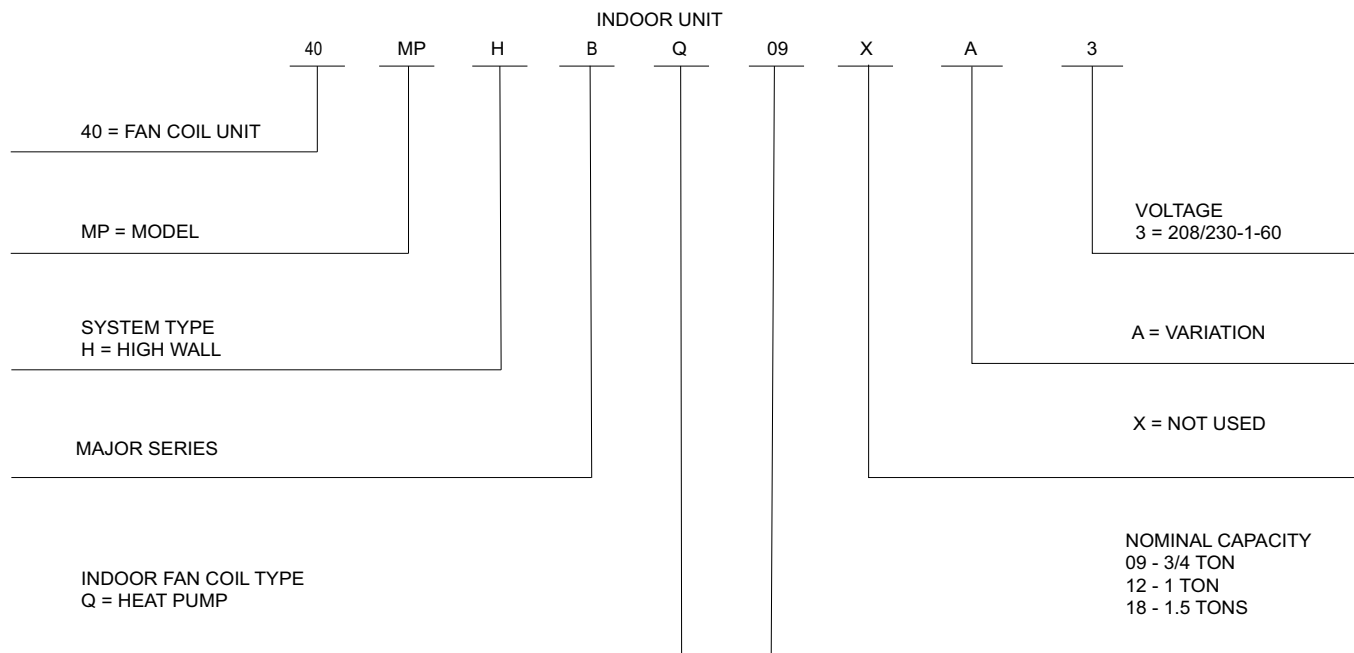
INTRODUCTION

This service manual provides the necessary information to service, repair, and maintain the indoor units. This manual has an appendix with data required to perform troubleshooting. Use the Table of Contents to locate a desired topic.

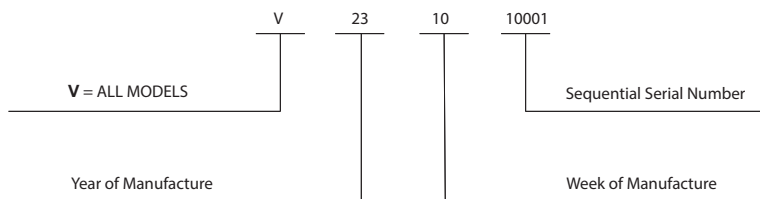
MODEL NUMBER NOMENCLATURE

Table 1 —Unit Sizes

SYSTEMS TONS	VOLTAGE/PH/HZ	INDOOR MODEL
9	208-230/1/60	40MPHBQ09XA3
12		40MPHBQ12XA3
18		40MPHBQ18XA3



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WIRING

All wires must be sized per NEC (National Electrical Code) or CEC (Canadian Electrical Code) and local codes. Use the Electrical Data table MCA (minimum circuit amps) and MOCP (maximum over current protection) to correctly size the wires and the disconnect fuse or breakers respectively.

Per the caution note, only stranded copper conductors with a 600 volt insulation rating wire must be used.

Recommended Connection Method for Power and Communication Wiring:

The main power is supplied to the outdoor unit. The field supplied 14/3 stranded wire with ground with a 600 volt insulation rating, power/communication wiring from the outdoor unit to indoor unit consists of four (4) wires and provides the power for the indoor unit.

Two wires are line voltage AC power, one is communication wiring (S) and the other is a ground wire. Wiring between the indoor and outdoor unit is polarity sensitive. The use of BX wire is NOT recommended.

If installed in a high Electromagnetic field (EMF) area and communication issues exists, a 14/2 stranded shielded wire can be used to replace L2 and (S) between the outdoor unit and indoor unit landing the shield onto ground in the outdoor unit only.



CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Wires should be sized based on NEC and local codes.



CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Be sure to comply with local codes while running wire from the indoor unit to the outdoor unit.

Every wire must be connected firmly. Loose wiring may cause the terminal to overheat or result in unit malfunction. A fire hazard may also exist. Ensure all wiring is tightly connected.

No wire should touch the refrigerant tubing, compressor or any moving parts.

Disconnecting means must be provided and shall be located within sight and readily accessible from the air conditioner.

Connecting cable with conduit shall be routed through the hole in the conduit panel.

CONNECTION DIAGRAM

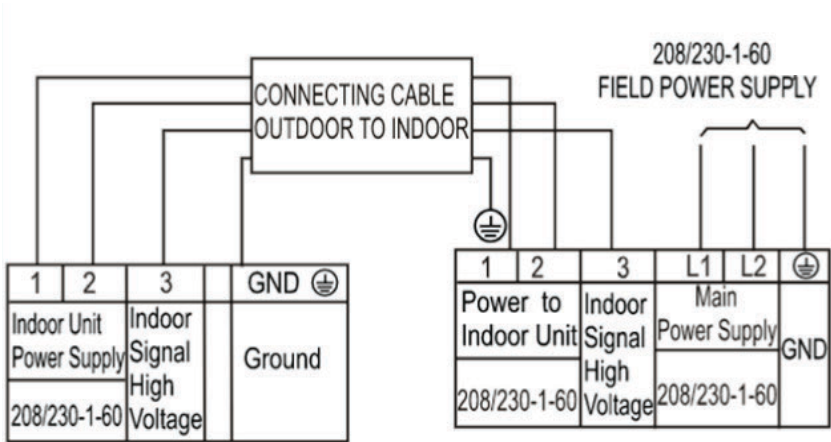


Fig. 1 — Connection Diagram 208-230V

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NOTES:

- 1. Do not use thermostat wire for any connection between the indoor and outdoor units.
- 2. All connections between the indoor and outdoor units must be as shown in Figure 1. The connections are polarity sensitive and **improper wiring will result in a fault code.**

WIRING DIAGRAMS

Sizes 09K to 12K

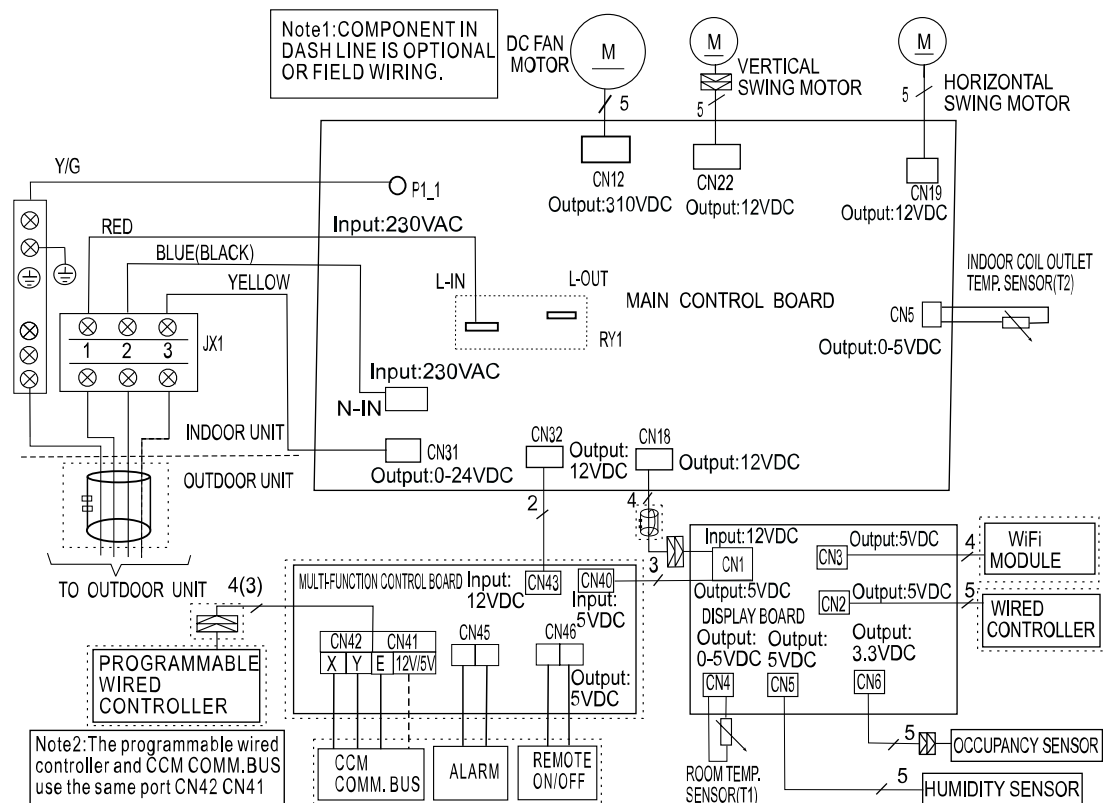


Fig. 2 —Wiring Diagram - Sizes 09K and 12K

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WIRING DIAGRAM

Size 18K

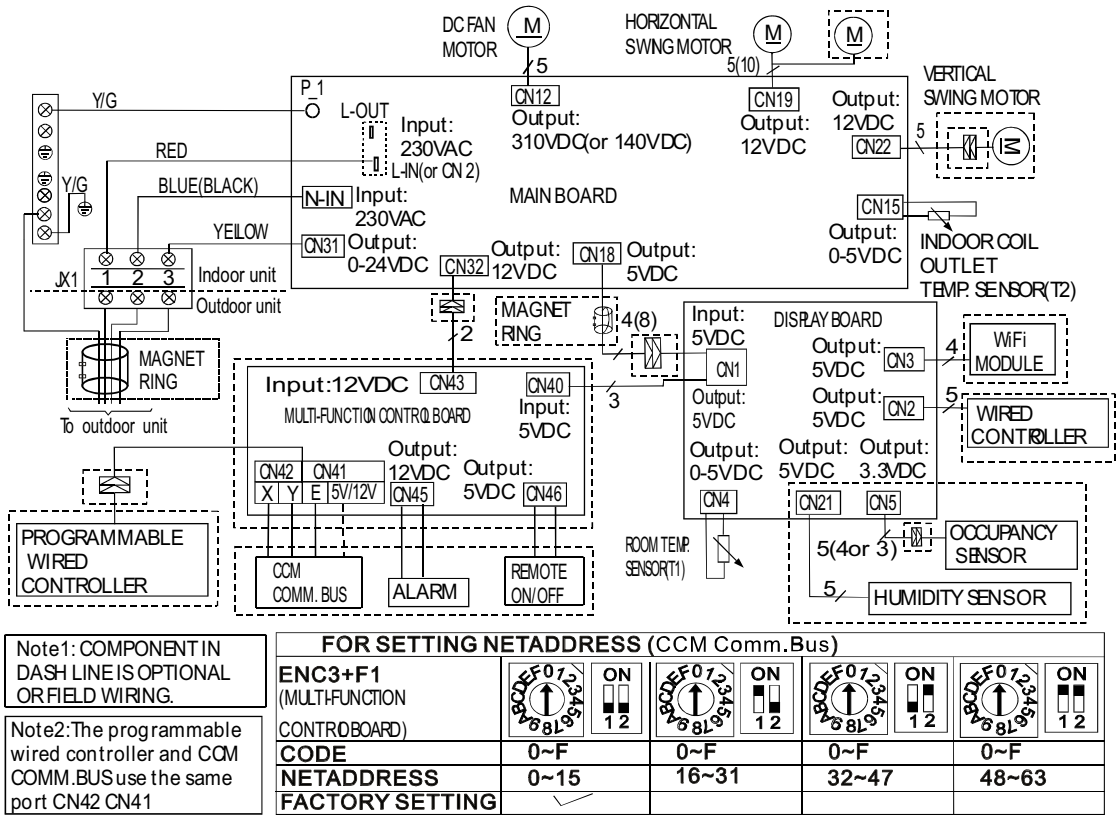
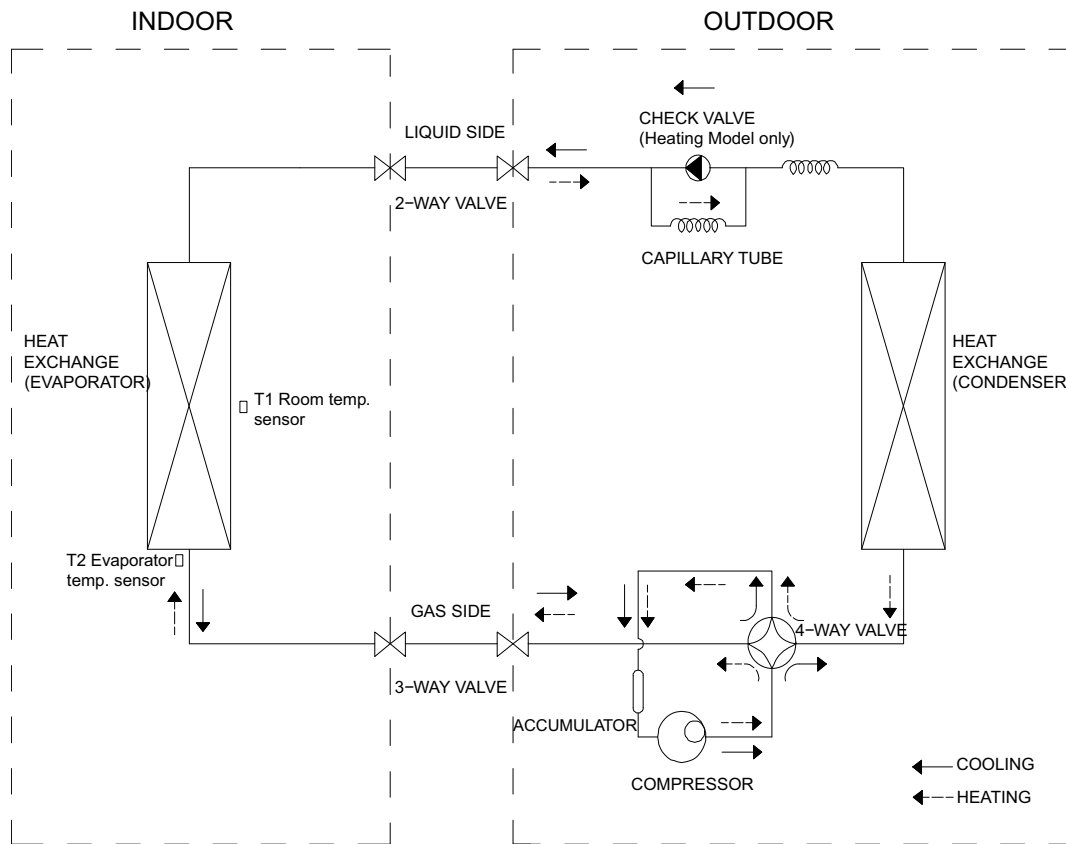


Fig. 3 — Wiring Diagram - Size 18K

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REFRIGERATION CYCLE DIAGRAM



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Fig. 4 — Heat Pumps

Refrigerant Lines

IMPORTANT: Both refrigerant lines must be insulated separately. Refer to the outdoor unit's installation instructions for other allowed piping lengths and refrigerant information.

SYSTEM EVACUATION AND CHARGING



CAUTION

UNIT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Never use the system compressor as a vacuum pump.

Refrigerant tubes and indoor coil should be evacuated using the recommended deep vacuum method of 500 microns. The alternate triple evacuation method may be used if the following procedure is followed. Always break a vacuum with dry nitrogen.

System Vacuum and Charge

Using Vacuum Pump

1. Completely tighten flare nuts on the line set at both the indoor and outdoor units. **DO NOT** open the service valves on the outdoor unit for the new installation or the replacement unit. Open the service valves on the outdoor unit if repairs have been made to the refrigerant sealed system. Connect the manifold gauge low pressure hose to the charge port of the gas side service valve (see Fig. 5).
2. Connect the charge hose to the vacuum pump.
3. Fully open the low pressure valve of manifold gauge (see Fig. 6).
4. Start the vacuum pump.
5. Evacuate using deep vacuum or triple evacuation method.
6. After evacuation is complete, fully close the pressure valve side of manifold gauge and stop the vacuum pump operation.
7. The factory charge contained in the outdoor unit is good for up to 25ft. (8 m) of line length. If vacuum is complete per Fig. 7 or 8, open service valves to release factory charge into the system.
8. Disconnect the charge hose from the charge connection of the gas side service valve.
9. Securely tighten the service valve caps.

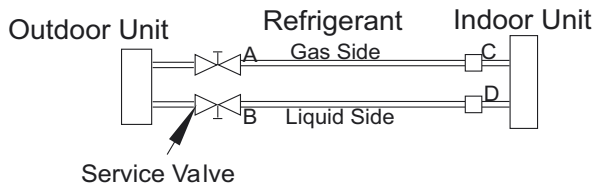


Fig. 5 — Service Valve

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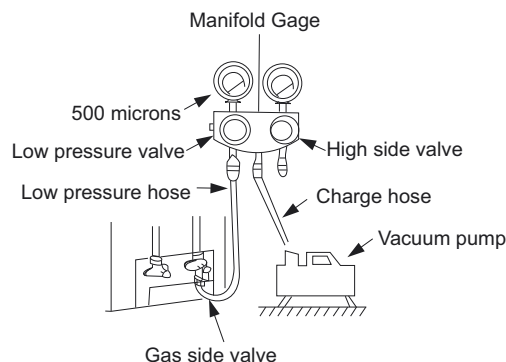


Fig. 6 — Manifold

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Deep Vacuum Method

The deep vacuum method requires a vacuum pump capable of pulling a vacuum of 500 microns and a vacuum gauge capable of accurately measuring this vacuum depth. The deep vacuum method is the most positive way of assuring a system is free of air and liquid water (see Fig. 7).

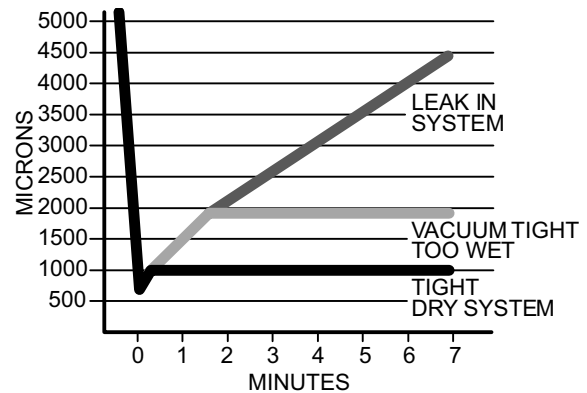


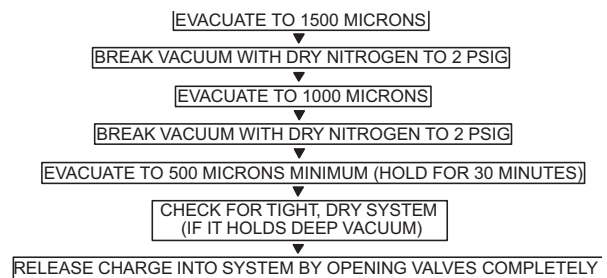
Fig. 7 — Deep Vacuum Graph

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Triple Vacuum Method

The triple evacuation method should be used. Refer to Fig. 8 and proceed as follows:

1. Pump the system down to 1500 microns and allow the pump to continue operating for an additional 15 minutes.
2. Close the service valves and shut off the vacuum pump.
3. Connect a dry nitrogen cylinder and regulator to the system and break vacuum until the system reaches 2 psig.
4. Close the service valve and allow the system to stand for 1hr. During this time, the dry nitrogen can diffuse throughout the system absorbing moisture.
5. Pump the system down to 1000 microns.
6. Break the vacuum with dry nitrogen (2 psig).
7. Pump the system down to 500 microns.
8. Perform the hold test for 30 minutes.



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Fig. 8 — Triple Evacuation Method

Final Tubing Check

IMPORTANT: Ensure that the factory tubing on both indoor and outdoor unit has not shifted during shipment. Ensure tubes are not rubbing against each other or any sheet metal. Pay close attention to feeder tubes, making sure wire ties on feeder tubes are secure and tight.

OPERATION MODES AND FUNCTIONS (9K-12K)

Abbreviation

Table 2 — Unit Element Abbreviations

ABBREVIATION	ELEMENT
T1	Indoor room temperature
T2	Evaporator Coil temperature
T3	Condenser Coil temperature
T4	Outdoor ambient temperature
Tsc	Adjusted setting temperature
TP	Compressor discharge temperature

Safety Features

Compressor Three-Minute Delay at Restart

Compressor functions are delayed for up to ten seconds upon the first start-up of the unit, and are delayed for up to three minutes upon subsequent unit restarts.

Automatic shutoff based on discharge temperature

If the compressor discharge temperature exceeds 226°F (108°C) for nine seconds, the compressor ceases operation.

Automatic shutoff based on fan speed

If the indoor fan speed registers below 300RPM or over 1500RPM for an extended period of time, the unit ceases operation and the corresponding error code is displayed on the indoor unit.

Inverter module protection

The inverter module has an automatic shutoff mechanism based on the unit's current, voltage, and temperature. If automatic shutoff is initiated, the corresponding error code is displayed on the indoor unit and the unit ceases operation.

Indoor fan delayed operation

- When the unit starts, the louver is automatically activated and the indoor fan begins to operate after a period of setting time or when the louver is in place.
- If the unit is in **HEATING** mode, the indoor fan is regulated by the anti-cold wind function.

Compressor Preheating

Preheating is automatically activated when the T4 sensor is lower than setting temperature.

Sensor redundancy and automatic shutoff

- If one temperature sensor malfunctions, the air conditioner continues operation and displays the corresponding error code, allowing for emergency use.
- When more than one temperature sensor malfunctions, the air conditioner ceases operation.

OPERATION MODES AND FUNCTIONS (9K-12K) (CONT)

Display Functions

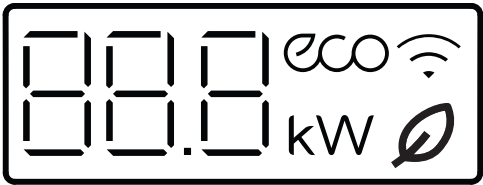


Fig. 9 — Unit Display Functions

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Table 3 — Unit Displays Function

FUNCTION	DISPLAY
Temperature	Set temperature value
Temperature (FAN and DRYING mode)	Room temperature
Activation of Timer ON, Fresh, Swing, Turbo, or Silent	ON(3S)
Cancellation of Timer OFF, Fresh, Swing, Turbo, or Silent	OF(3S)
Defrost	dF
Self-clean	SC
Heating in room temperature under 46°F (8°C) or 54°F (12°C)	FP
Fresh (Not available on these systems)	
ECO function	
Wi-Fi Control	
The current operation power (Not available on these systems)	kW

FAN Mode

When the **FAN** mode is activated:

- The outdoor fan and compressor stop.
- Temperature control is disabled and the indoor room temperature is displayed.
- The indoor fan speed can be set to 1%~100%, or **AUTO**.
- The louver operations are identical to those in **COOLING** mode.
- Auto fan: In **FAN-ONLY** mode, the **AC** operates the same as auto fan in the **COOLING** mode with the temperature set at 75°F (24°C). (Tsc=75°F (24°C)).

COOLING Mode

Compressor control

Reach the configured temperature:

1. When the compressor runs continuously for less than 120 minutes.
 - If the following conditions are satisfied, the compressor ceases operation.
 - While the calculated frequency (fb) is less than the minimum limit frequency (FminC).
 - While protective time is more than or equal to ten minutes.
 - While T1 is lower than or equal to (Tsc-CDIFTEMP -1°F (-0.5°C))

NOTE: CDIFTEMP is the EEPROM setting parameter. It is 4°F (2°C).

2. When the compressor runs continuously for more than 120 minutes.
 - If the following conditions are satisfied, the compressor ceases operation.
 - When calculated frequency (fb) is less than minimum limit frequency (FminC).
 - When protective time is more than or equal to ten minutes.
 - When T1 is lower than or equal to (Tsc-CDIFTEMP).

NOTE: CDIFTEMP is the EEPROM setting parameter. It is 4°F (2°C).

3. If one of the following conditions is satisfied, regardless of time.
 - Compressor running frequency is more than the test frequency.
 - When the compressor running frequency is equal to the test frequency, T4 is more than 59°F (15°C) or no T4 or T4 fault.
 - Change setting temperature
 - Turbo or sleep function on/off
 - Various frequency limit shutdown occurs

NOTE: CDIFTEMP is EEPROM setting parameter. It is 4°F (2°C).

Indoor Fan Control

1. In the **COOLING** mode, the indoor fan operates continuously. The fan speed can be set to 1%-100%, or **AUTO**.
2. **AUTO** fan
 - **Descent Curve**
 - When T1-Tsc is lower than or equal to 6°F (3.5°C), fan speed reduces to 80%;
 - When T1-Tsc is lower than or equal to 2°F (1°C), fan speed reduces to 60%;
 - When T1-Tsc is lower than or equal to 1°F (0.5°C), fan speed reduces to 40%;
 - When T1-Tsc is lower than or equal to 0°F (0°C), fan speed reduces to 20%;
 - When T1-Tsc is lower than or equal to -1°F (-0.5°C), fan speed reduces to 1%.
 - **Rise Curve**
 - When T1-Tsc is higher than 0°F (0°C), fan speed increases to 20%;
 - When T1-Tsc is higher than 1°F (0.5°C), fan speed increases to 40%;
 - When T1-Tsc is higher than 2°F (1°C), fan speed increases to 60%;
 - When T1-Tsc is higher than 3°F (1.5°C), fan speed increases to 80%;
 - When T1-Tsc is higher than 7°F (4°C), fan speed increases to 100%.

Outdoor Fan Control

- The outdoor unit runs at a different fan speed according to T4 and the compressor running frequency.
- For different outdoor units, the fan speeds are different.

Condenser Temperature Protection

When the condenser temperature exceeds a configured value, the compressor ceases operations.

Evaporator Temperature Protection

When the evaporator temperature drops below a configured value, the compressor and outdoor fan ceases operations.

HEATING Mode

Compressor control

1. Reach the configured temperature:
 - If the following conditions are satisfied, the compressor ceases operation.
 - While the calculated frequency (fb) is less than the minimum limit frequency (FminC).
 - When the protective time is more than or equal to ten minutes.
 - When T1 is higher than or equal to Tsc+HDIFTEMP2.

NOTE: HDIFTEMP2 is the EEPROM setting parameter. It is 4°F (2°C).

- If one of the following conditions is satisfied, regardless of time.
 - Compressor running frequency is more than test frequency.
 - When the compressor running frequency is equal to the test frequency, T4 is more than 59°F (15°C) or no T4 or T4 fault.
 - Change the setting temperature.
 - **TURBO** or **SLEEP** function on or off.
- 2. When the current is higher than the predefined safe value, the surge protection is activated, causing the compressor to cease operations.

Indoor Fan Control

1. In the **HEATING** mode, the indoor fan operates continuously. The fan speed can be set to 1%-100%, or muted.
2. **AUTO** fan
 - **Rise curve**
 - When T1-Tsc is higher than -3°F (-1.5°C), fan speed reduces to 80%;
 - When T1-Tsc is higher than 0°F (0°C), fan speed reduces to 60%;
 - When T1-Tsc is higher than 1°F (0.5°C), fan speed reduces to 40%;
 - When T1-Tsc is higher than 2°F (1°C), fan speed reduces to 20%.
 - **Descent curve**
 - When T1-Tsc is lower than or equal to 1°F (0.5°C), fan speed increases to 20%;
 - When T1-Tsc is lower than or equal to 0°F (0°C), fan speed increases to 60%;
 - When T1-Tsc is lower than or equal to -3°F (-1.5°C), fan speed increases to 80%;
 - When T1-Tsc is lower than or equal to -1.5°F (-3°C), fan speed increases to 100%.

Outdoor Fan Control

- The outdoor unit runs at a different fan speed according to T4 and compressor running frequency.
- For different outdoor units, the fan speeds differ.

DEFROSTING Mode

- The unit enters defrosting mode according to changes in the temperature value of T3, T4 as well as the compressor running time.
- In the **DEFROSTING** mode, the compressor continues to run, the indoor and outdoor motor will cease operation, the defrost light of the indoor unit will turn on, and the "df" symbol appears.
- If any one of the following conditions is satisfied, defrosting ends and the machine switches to the normal **HEATING** mode:
 - T3 rises above TCDE1C.
 - T3 maintained above TCDE2C for 80 seconds.
 - Unit runs for 15 minutes consecutively in the **DEFROSTING** mode.

Evaporator Temperature Protection

- **Off:** Compressor stops.
- **Decrease:** Decrease the running frequency to the lower level per 20 seconds.
- **Hold:** Keep the current frequency.
- **Resume:** No limitation for frequency.

AUTO Mode

- This mode can be selected with the remote controller and the setting temperature can be changed between 61°F~86°F (16°C~30°C).
- In the **AUTO** mode, the machine selects the **COOLING**, **HEATING**, **AUTO-DRYING** or **FAN-ONLY** mode on the basis of T1, Ts, T4 and relative humidity.
- If the setting temperature is modified, the machine selects a new running function.

DRY Mode

In the **DRY** mode, the air conditioner operates the same as auto fan in the **COOLING** mode.

1. Mute function is active.
 - All protections are activated and operate the same as they do that in **COOLING** mode.
2. Low Room Temperature Protection
 - If the room temperature is lower than 50°F (10°C), the compressor ceases operations and does not resume until the room temperature exceeds 54°F (12°C).

FORCED OPERATION Function

• Forced COOLING Mode

The compressor and outdoor fan continues to run and the indoor fan runs at rated speed. After running for 30 minutes, the air conditioner switches to **AUTO** mode with a preset temperature of 75°F (24°C).

• Forced AUTO Mode:

Forced AUTO Mode operates the same as normal **AUTO** mode with a preset temperature of 75°F (24°C).

- The unit exits the forced operation when it receives the following signals:
 - Switch on
 - Switch off
 - Timer on
 - Timer off
 - Changes in:
 - Mode
 - Fan Speed
 - Setting Temperature

TIMER Function

- The Timing range is 24 hours.
- **Timer on.** The machine turns on automatically when reaching the setting time.
- **Timer off.** The machine turns off automatically when reaching the setting time.
- **Timer on/off.** The machine turns on automatically when reaching the setting “on” time, and then turns off automatically when reaching the setting “off” time.
- **Timer off/on.** The machine turns off automatically when reaching the setting “off” time, and then turns on automatically when reaching the setting “on” time.
- The timer function does not change the AC current operation mode. Suppose the AC is off now, it will not start up first after setting the “timer off” function. And when reaching the setting time, the timer LED turns off and the AC running mode does not change.
- The setting time is relative time.
- The AC exits the timer function when it has malfunction.

SLEEP Function

- The **SLEEP** function is available in **COOLING**, **HEATING**, or **AUTO** modes.
- The operational process for sleep mode is as follows:
 - When cooling, the temperature rises 2°F (1°C) (to not higher than 86°F (30°C)) every hour. After 2 hours, the temperature stops rising and the indoor fan is fixed at low speed.
 - When heating, the temperature decreases 2°F (1°C) (to not lower than 61°F (16°C)) every hour.
 - After 2 hours, the temperature stops decreasing and the indoor fan is fixed at low speed. Anti-cold wind function takes priority.
- The operating time for the **SLEEP** mode is 8 hours, after which, the unit exits this mode and turns off.
- The timer setting is available in this mode.

AUTO-RESTART Function

- The indoor unit has an auto-restart module that allows the unit to restart automatically. The module automatically stores the current settings and, in the case of a sudden power failure, will restore those settings automatically within 3 minutes after power returns.
- If there is a power failure while the unit is running, the compressor starts 3 minutes after the unit restarts. If the unit was already off before the power failure, the unit stands by.

46°F (8°C) Heating

In the **HEATING** mode, the temperature can be set to as low as 46°F (8°C), preventing the indoor area from freezing if unoccupied during severe cold weather.

ECO Function

- Use to enter the energy efficient mode.
 - Under the **COOLING** mode, press **ECO**, the remote controller adjusts the temperature automatically to 75°F (24°C), **AUTO** fan speed to save energy (however only if the set temperature is less than 75°F (24°C)). If the set temperature is more than 75°F (24°C) and 86°F (30°C), press **ECO**, the fan speed changes to **AUTO**, the set temperature remains unchanged.
- When AC receives signals, such as switch off, **Turbo** operation, **Silence** operation, **Self Clean** operation, Forced **COOLING** operation, mode setting, **SLEEPING** mode, or adjusting the set temperature to less than 75°F (24°C), the AC exits the **ECO** operation.
- Operation time in **ECO** mode is 8 hours. After 8 hours the air conditioner exits this mode.
- If there is a malfunctioning temperature sensor in, the air conditioner exits **ECO** mode.
- The indoor fan runs at auto fan when it enters the **ECO** mode. The setting temperature and setting fan speed can be changed with the remote controller.

SELF CLEAN

- Press **Self Clean** when the unit is in the **COOLING** or **DRYING** mode, the indoor unit runs at the low fan speed for 16 minutes then turns off.
- **Self Clean** keeps the indoor unit dry and prevents mold growth.

FOLLOW ME

- If you press **Follow Me** on the remote controller, the indoor unit beeps; this indicates the **Follow Me** function is active.
- Once active, the remote controller sends a signal every 3 minutes, with no beeps. The unit automatically sets the temperature according to the measurements from the remote controller.
- The unit only change modes if the information from the remote controller makes it necessary, not from the unit's temperature setting.
- If the unit does not receive a signal for 7 minutes or you press **Follow Me**, the function turns off. The unit regulates temperature based on its own sensor and settings.

SILENCE

Press **SILENCE** on the remote controller to enable the **SILENCE** function. While this function is active, the indoor unit runs at faint breeze (1% fan speed), which reduces noise to the lowest possible level.

OCCUPANCY SENSOR (Sizes 09K-12K)

With the built-in infrared sensor, the indoor unit detects human movement. The compressor operates in a low frequency if you leave the room for 30 minutes. The compressor operates in a lower frequency if you leave the room for 120 minutes, and resumes automatically when you come back, which helps save energy.

TROUBLESHOOTING

Safety


WARNING

UNIT DAMAGE HAZARD

Electricity remains in capacitors even when the power supply is off.
Ensure the capacitors are fully discharged before troubleshooting.

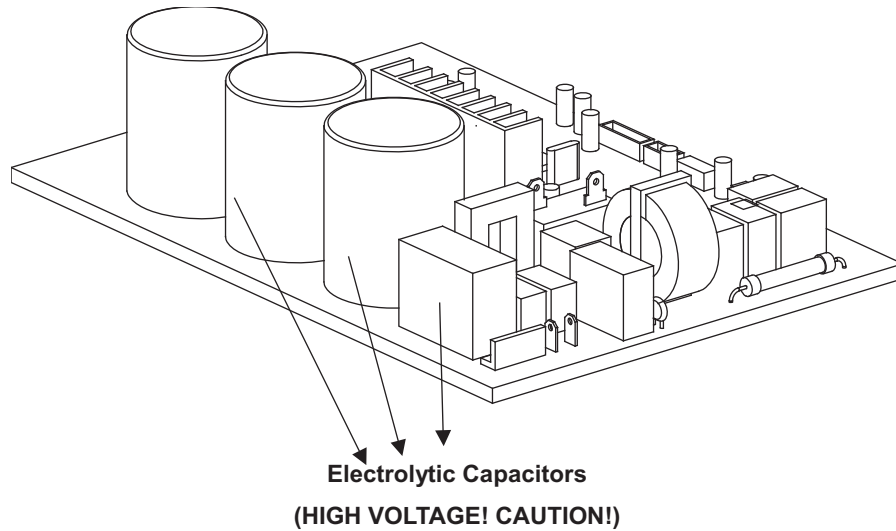


Fig. 10 —Electrolytic Capacitors

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For other models, please connect discharge resistance (approximately 100Ω 40W) or a soldering iron (plug) between the +, - terminals of the electrolytic capacitor on the other side of the outdoor PCB.

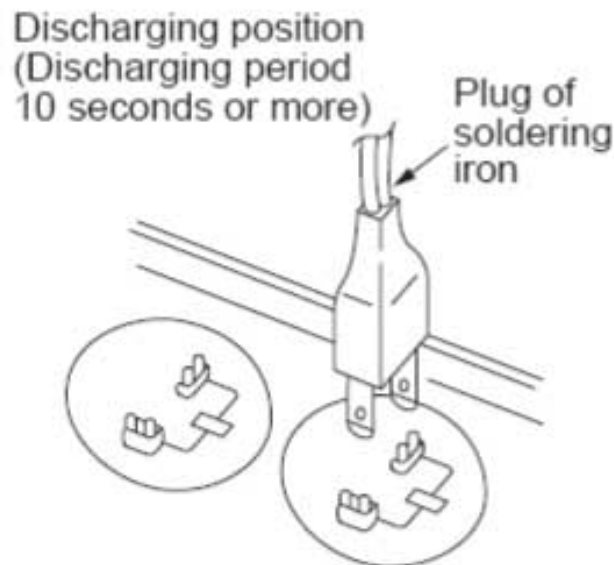


Fig. 11 — Discharge Position

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NOTE: Fig. 11 is for reference only. Actual appearances may vary.

ERROR DIAGNOSIS AND TROUBLESHOOTING WITHOUT ERROR CODE


WARNING

UNIT DAMAGE HAZARD

Be sure to turn off unit before any maintenance to prevent damage or injury.

Table 4 — Remote Maintenance

Remote Maintenance	Electrical Circuit							Refrigerant Circuit							Others					
Possible causes of trouble	Power failure	The main power tripped	Loose connections	Faulty transformer	The voltage is too high or too low	The remote control is powered off	Broken remote control	Dirty air filter	Dirty condenser fins	The setting temp is higher/lower than the room's (cooling/heating)	The ambient temp. is too high/low when the mode is cooling/heating	Fan mode	SILENCE function is activated (optional function)	Frosting and defrosting frequently	Heavy load condition	Loose hold down bolts and/or screws	Not air tight	Air inlet or outlet of either unit is blocked	Interference from cell phones towers and remote boosters	Shipping plates remained attached
Unit will not start	☆	☆	☆	☆																
Operation is erratic, unpredictable or unit is unresponsive																		☆		
Cannot set desired temp.						☆	☆													
Unit is on however the wind is not cold (hot)										☆	☆	☆								
Unit runs, then shortly stops					☆					☆	☆									
The unit starts up and stops often					☆						☆			☆				☆		
Unit runs continuously however insufficient cooling (heating)								☆	☆	☆	☆		☆		☆		☆	☆		
Cool can not change to heat																				
Unit is noisy																☆				☆
Unit emits bad odor								☆												
Test method/ remedy	Test voltage	Close the power switch	Inspect connections - tighten	Change the transformer	Test voltage	Replace the battery of the remote control	Replace remote control	Clean or replace	Clean	Adjust the setting temp.	Turn the AC later	Adjust to Cool mode	Turn off SILENCE function.	Turn the AC later	Check the heat load	Tighten bolts or screws	Close all windows and doors	Remove the obstacles	Reconnect the power or press ON/OFF button on remote control to restart	Remove them

ERROR DIAGNOSIS AND TROUBLESHOOTING WITHOUT ERROR CODE (continued)

Table 5 — Field Maintenance

Field Maintenance	Electrical Circuit														
Possible causes of trouble	Power failure	Blown fuse or varistor	Loose connections	Shorted or broken wires	Safety device opens	Faulty thermostat/room temp sensor	Wrong setting place of temp sensor	Faulty transformer	Shorted or open capacitor	Faulty magnetic contact for compressor	Faulty magnetic contact for fan	Low voltage	Faulty stepping motor	Shorted or grounded compressor	Shorted or grounded fan motor
Unit will not start	☆	☆	☆	☆	☆			☆							
Compressor will not start however the fan runs				☆		☆			☆	☆				☆	
Compressor and condenser (outdoor) fan will not start				☆		☆				☆					
Evaporator (indoor) fan will not start				☆					☆		☆				☆
Condenser (outdoor) fan will not start				☆		☆			☆		☆				☆
Unit runs but shortly stops										☆		☆			
Compress or short cycles due to overload										☆		☆			
High discharge pressure															
Low discharge pressure															
High suction pressure															
Low suction pressure															
Unit runs continuously but insufficient cooling															
Too cool						☆	☆								
Compressor is noisy															
Horizontal louver can not revolve			☆	☆									☆		
Test method / remedy	Test voltage	Inspect fuse type & size	Inspect connections - tighten	Test circuits with tester	Test continuity of safety device	Test continuity of thermostat / sensor & wiring	Place the temp. sensor at the central of air inlet grille	Check control circuit with tester	Check capacitor with tester	Test continuity of coil & contacts	Test continuity of coil & contacts	Test voltage	Replace the stepping motor	Check resistance with Megger tester	Check resistance with Megger tester

ERROR DIAGNOSIS AND TROUBLESHOOTING WITHOUT ERROR CODE (continued)

Table 6 — Field Maintenance

Field Maintenance	Refrigerant Circuit																	
Possible causes of trouble	Compressor stuck	Refrigerant Shortage	Restricted liquid line	Dry air filter	Dirty evaporator coil	Insufficient air through evaporator coil	Refrigerant overcharge	Dirty or partially blocked condenser	Air or non-condensable gas in refrigerant cycle	Short cycling of condensing air	High temp. condensing medium	Insufficient condensing medium	Broken compressor internal parts	Insufficient compressor	Expansion valve obstructed	Expansion valve or capillary tube closed completely	Leaking power element on expansion valve	Power installation of feeler bulb
Unit will not start																		
Compressor will not start however the fan runs	☆																	
Compressor and condenser (outdoor) fan will not start																		
Evaporator (indoor) fan will not start																		
Condenser (outdoor) fan will not start																		
Unit runs but shortly stops		☆	☆				☆	☆								☆	☆	
Compress or short cycles due to overload		☆					☆	☆										
High discharge pressure							☆	☆	☆	☆	☆	☆						
Low discharge pressure		☆												☆				
High suction pressure							☆							☆				☆
Low suction pressure		☆	☆	☆	☆	☆									☆	☆	☆	
Unit runs continuously but insufficient cooling		☆	☆	☆	☆	☆		☆	☆	☆				☆				
Too cool																		
Compressor is noisy							☆						☆					
Horizontal louver can not revolve																		
Test method / remedy	Replace the compressor	Leak test	Replace the restricted part	Clean or replace	Clean coil	Check fan	Change charged refrigerant volume	Clean condenser or remove obstacle	Purge, evacuate and recharge	Remove obstruction to air flow	Remove obstruction in air or water flow	Remove obstruction in air or water flow	Replace compressor	Test compressor efficiency	Replace valve	Replace valve	Replace valve	Fix feeler bulb

ERROR DIAGNOSIS AND TROUBLESHOOTING WITHOUT ERROR CODE (continued)

Table 7 — Field Maintenance

Field Maintenance	Others				
Possible causes of trouble	Heavy load condition	Loosen hold down bolts and/or screws	Shipping plates remain attached	Poor choices of capacity	Contact of piping with other piping or external plate
Unit will not start					
Compressor will not start however the fan runs					
Compressor and condenser (outdoor) fan will not start					
Evaporator (indoor) fan will not start					
Condenser (outdoor) fan will not start					
Unit runs but shortly stops					
Compress or short cycles due to overload					
High discharge pressure					
Low discharge pressure					
High suction pressure	☆				
Low suction pressure					
Unit runs continuously but insufficient cooling	☆			☆	
Too cool					
Compressor is noisy		☆	☆		☆
Horizontal louver can not revolve					
Test method / remedy	Check heat load	Tighten bolts or screws	Remove them	Choose AC of larger capacity or add the number of units	Fix piping as not to touch each other or external pipe

QUICK MAINTENANCE BY ERROR CODE

If you do not have the time to test whether specific parts are faulty, you can directly change the required parts according the error code.

The parts to replace by error code are listed in Tables 8 and 9.

Table 8 — Error Codes

PART REQUIRING REPLACEMENT	ERROR CODE									
	EH 00/ EH 0A	EL 01	EH 02	EH 03	EH 60	EH 61	EH 0B	EL 0C	EC 56	PC 08
Indoor PCB	√	√	√	√	√	√	√	√	x	x
Outdoor PCB	x	√	x	x	x	x	x	x	√	√
Display board	x	x	x	x	x	x	√	x	x	x
Indoor fan motor	x	x	x	√	x	x	x	x	x	x
T1 sensor	x	x	x	x	√	x	x	x	x	x
T2 Sensor	x	x	x	x	x	√	x	√	x	x
T2B Sensor	x	x	x	x	x	x	x	x	√	x
Reactor	x	√	x	x	x	x	x	x	x	x
Compressor	x	x	x	x	x	x	x	x	x	√
Additional refrigerant	x	x	x	x	x	x	x	√	x	x

Table 9 — Error Codes

PART REQUIRING REPLACEMENT	EC 53	EC 52	EC 54	EC 51	EC 07	PC 00	PC 01	PC 02	PC 03	PC 04
Outdoor PCB	√	√	√	√	√	√	√	√	√	√
Indoor fan motor	x	x	x	x	x	x	x	x	x	x
Outdoor fan motor	x	x	x	x	√	√	x	√	x	√
T3 Sensor	x	√	x	x	x	x	x	x	x	x
T4 Sensor	√	x	x	x	x	x	x	x	x	x
TP Sensor	x	x	√	x	x	x	x	x	x	x
Reactor	x	x	x	x	x	x		x	x	x
Compressor	x	x	x	x	x	√	x	x	x	√
IPM module board	x	x	x	x	x	√	√	√	x	√
High pressure protector	x	x	x	x	x	x	x	√	x	x
Low pressure protector	x	x	x	x	x	x	x	x	√	x
Additional refrigerant	x	x	x	x	x	x	x	x	√	x

TROUBLESHOOTING BY ERROR CODE

Common Check Procedures

Temperature Sensor Check

Disconnect the temperature sensor from PCB, measure the resistance value with a tester.

Temperature Sensors.

Room temp. (T1) sensor,

Indoor coil temp. (T2) sensor,

Outdoor coil temp.(T3) sensor,

Outdoor ambient temp.(T4) sensor,

Compressor discharge temp.(Tp) sensor.

Measure the resistance value of each winding with a multi-meter.



A221379

Fig. 12 — IPM Continuity Check

IPM Continuity Check

Turn off the power. Allow the large capacity electrolytic capacitors to discharge completely, and dismount the IPM. Use a digital tester to measure the resistance between P and UVWN; UVW and N.

Table 10 — IPM Continuity Check

Digital Tester		Normal Resistance Value	Digital Tester		Normal Resistance Value
(+) Red	(-) Black	∞ (Several M Ω)	(+) Red	(-) Black	∞ (Several M Ω)
P	N		U	N	
	U		V		
	V		W		
	W		(+) Red		

OPERATION MODES AND FUNCTIONS (18K)

FAN Mode

1. Outdoor fan and compressor stop
2. Temperature setting function is disabled and no setting temperature appears.
3. Indoor fan can be set to high/med/low/auto
4. The louver operates the same as in the **COOLING** mode.

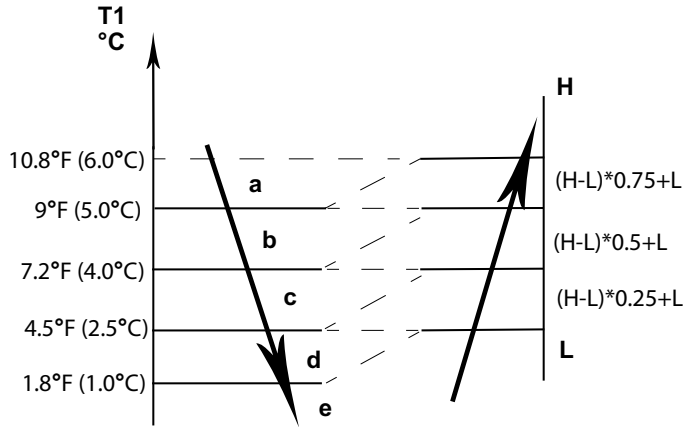


Fig. 13 — AUTO FAN Mode

A221402

COOLING Mode

Indoor Fan Running Rules:

In the **COOLING** mode, the indoor fan runs all the time and the speed can be selected as **HIGH**, **MEDIUM**, **LOW** and **AUTO**. When the setting temperature is reached, if the compressor stops running, the indoor fan motor runs at the minimum or setting speed. The indoor fan is controlled by the rules shown in Fig. 14.

Setting fan speed	T1-Td °C(°F)	Actual fan speed
H	A	H+ (H+=H+G)
	B	H (=H)
	C	H- (H-=H-G)
M	D	M+ (M+=M+Z)
	E	M (M=M)
	F	M- (M-=M-Z)
L	G	L+ (L+=L+D)
	H	L (L=L)
	I	L- (L-=L-D)

Fig. 14 — Indoor Fan Running Rules

A221403

Evaporator Temperature Protection

When the evaporator temperature is less than the setting value, the compressor stops.

HEATING Mode

Indoor Fan Running Rules:

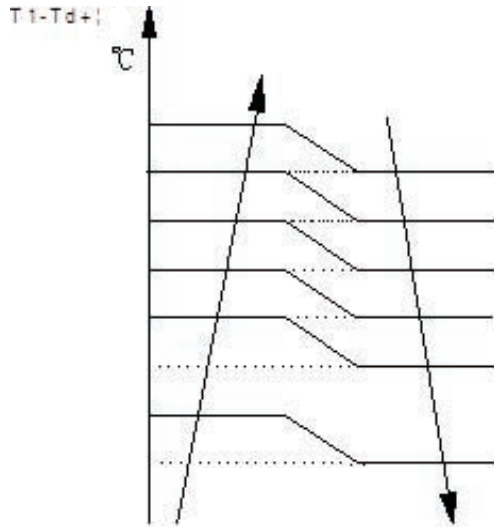
When the compressor is on, the indoor fan can be set to **HIGH**, **MEDIUM**, **LOW**, **AUTO**, **MUTE**. When the indoor unit coil temperature is low, the anti-cold air function starts and the indoor fan motor runs at a low speed and the speed cannot be changed. When the temperature is lower than the setting value, the indoor fan motor stops.

When the indoor temperature reaches the setting temperature, the compressor stops and the indoor fan motor runs at the minimum speed or setting speed. The anti-cold air function is valid. The indoor fan is controlled as shown in Fig. 15.

Setting fan speed	T1-Td °C	Actual fan speed
H		H- (H-=H-G)
		H (=H)
		H+ (H+=H+G)
M		M- (M-=M-Z)
		M (M=M)
		M+ (M+=M+Z)
L		L- (L-=L-D)
		L (L=L)
		L+ (L+=L+D)

Fig. 15 — Indoor Fan Running Rules

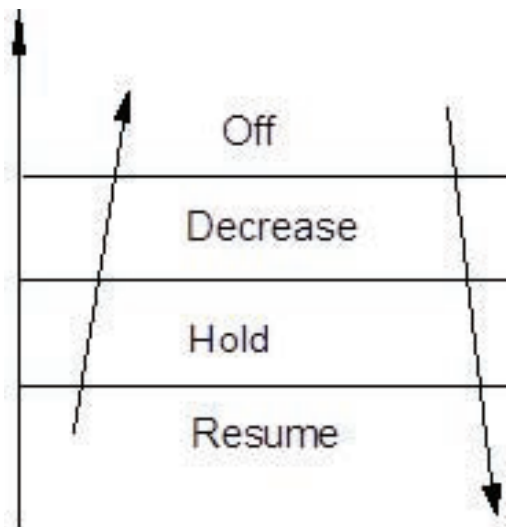
A221405

AUTO Fan Action in HEATING Mode

A221406

Fig. 16 — AUTO Fan Action in HEATING Mode**DEFROSTING Mode**

The air conditioner enters the **DEFROSTING** mode according to the T3 temperature value and the T3 temperature changes the value range plus the compressor's running time. During the **DEFROSTING** mode, the compressor continues to run, the indoor and outdoor motors stop, the indoor unit defrost lamp illuminates and "dF" appears.

Evaporator Coil Temperature Protection

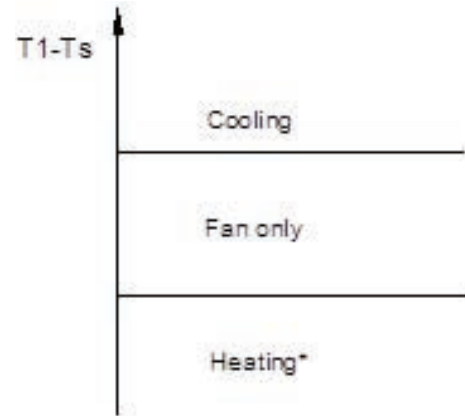
A221407

Fig. 17 — Evaporator Coil Temperature Protection

When the evaporator temperature is higher than the setting protection value, the compressor stops.

AUTO Mode

AUTO mode can be chosen with the remote controller and the setting temperature can be changed between 62.6°F(17°C)~86°F(30°C). In **AUTO** mode, the air conditioner chooses the **COOLING**, **HEATING** or **FAN ONLY** mode according to DT (DT = T1-Ts).



A221408

Fig. 18 — AUTO Mode

The indoor fan runs under the **AUTO** fan in the relevant mode. The louver operates the same as in the relevant mode. If the air conditioner switches between the **HEATING** and **COOLING** mode, the compressor stops for a certain period of time and then chooses the mode according to T1-Ts. If the setting temperature is modified, the air conditioner chooses the running function again.

DRYING Mode**Indoor Fan Speed is Fixed**

The Indoor fan speed is fixed at **BREEZE** and can not be changed. The louver angle is the same as in the **COOLING** mode.

Low Indoor Room Temperature Protection

In the **DRYING** mode, if the room temperature is lower than 50°F (10°C), the compressor stops and will not resume until the room temperature exceeds 53.6°F (12°C).

Evaporator Anti-Freezing Protection

The evaporator anti-freezing protection condenser high temperature protection and outdoor unit frequency limit are active and the same when in the **COOLING** mode.

Outdoor Fan

The outdoor fan operates the same as in the **COOLING** mode.

Forced Operation Function

When the air conditioner is off, press **TOUCH** to engage the **Forced AUTO** mode. Press **TOUCH** again within 5 seconds to engage the **Forced COOLING** mode. In the **Forced AUTO**, **Forced COOLING** or any other operation mode, press **TOUCH** to turn off the air conditioner.

Forced Operation Mode

In the **Forced OPERATION** mode, all the general protections and the remote controller are available.

Operation Rules

Forced Cooling Mode

The compressor runs at the F2 frequency and the indoor fan runs in the **BREEZE** mode. After running for 30 minutes, the air conditioner enters the **AUTO** mode at the 75.2°F (24°C) setting temperature.

Forced Auto Mode:

The **Forced AUTO** mode is the same as the normal **AUTO** mode with a 75.2°F (24°C) setting temperature.

Forced DEFROSTING Mode:

1. Press and hold **AUTO/COOL** for 5s to enter the mode. The indoor fan stops and the defrosting lamp **df** illuminates. Use the remote controller to exit this mode and turn off the air conditioner to stop the normal **DEFROSTING** mode.
2. To exit the **Forced DEFROSTING** mode, press and hold **AUTO/COOL** for 5s again.

AUTO-RESTART Function

The indoor unit is equipped with the **AUTO-RESTART** function, which is carried out through an auto-restart module. In the event of a sudden power failure, the module memorizes the setting conditions prior to the power failure. The air conditioner resumes the previous operation setting (not including the **SWING** function) automatically three (3) minutes after power is reestablished.

If the memorization condition is the **Forced COOLING** mode, the air conditioner runs in the **COOLING** mode for 30 minutes and turns to the **AUTO** mode at the 75.2°F(24°C) setting temperature. If the air conditioner is off before the power turns off and the air conditioner is required to start up, the compressor delays start-up for 1 minute before powering on. In other instances, the compressor waits three (3) minutes before restarts.

Refrigerant Leakage Detection

With this new technology, the display area displays “**EC**” when the outdoor unit detects a refrigerant leak. This function is only active in the **COOLING** mode. The function can further prevent the compressor from being damaged by a refrigerant leak or a compressor overload.

- **Open Condition:** When the compressor is active, the value of the coil temperature of evaporator T2 experiences no to very little change.

Louver Position Memory Function

When starting the air conditioner again after a shut down, the louver returns to the angle originally set by the user, however the precondition is that the angle must be within the allowable range. If the louver exceeds the allowable range, the air conditioner memorizes the maximum angle of the louver. During operation, if the power fails or the user shuts down the air conditioner in the **TURBO** mode, the louver returns to the default angle.

46°F (8°C) Heating

When the compressor is running, the indoor fan motor runs without the **ANTI-COLD** air function. When the compressor is off, the indoor fan motor is off.

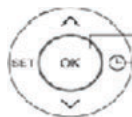
Silence Operation

Press **SILENCE** on the remote controller to initiate the **SILENCE** function. When **SILENCE** is activated, the compressor running frequency remains lower than **F2** and the indoor unit emits a faint breeze, which reduces the noise to the lowest level and creates a quiet and comfortable room for the user.

Inquiry Mode

To enter Inquiry Mode:

Press and hold together the **On/Off** and **Fan** buttons   for 8 seconds. The remote control remains in **Inquiry Mode** for 1 minute if no button is pressed. In the **Inquiry Mode**, the remote display cancels all icons except **AUTO**, **COOL**, **DRY**, **HEAT** and battery strength. The digital display defaults to “0” upon entering the **Inquiry Mode**. In **Inquiry Mode**, each digital code (from 0 to 30) is accessed by pressing the **UP** or **DOWN** arrow.



A221410

The Inquiry information appears on the high wall indoor unit display in approximately 1 second of accessing the digital code. Press **OK** to send as well.

Table 11 — Inquiry Codes and Symbols

CODE	INQUIRY SYMBOL	DESCRIPTION
Code 0		None
Code 1	T1	Indoor ambient
Code 2	T2	Indoor pipe
Code 3	T3	Outdoor pipe
Code 4	T4	Outdoor air
Code 5	TP (T5)	Compressor discharge
Code 6	FT	Compressor target frequency
Code 7	Fr	Compressor run frequency
Code 8	dL	Unit amperage
Code 9	Uo	Unit voltage
Code 10	Sn	Capacity test (special usage)
Code 11	----	N/A
Code 12	Pr	Indoor fan speed
Code 13	Lr	Electronic Expansion Valve (EEV) opening
Code 14	ir	Indoor fan speed
Code 15	HU	Humidity
Code 16	TT	Set point compensation temperature
Code 17	dT	Dust concentration (not used)
Code 18	WIFI	Wi-Fi signal strength
Code 19	----	N/A
Code 20	oT	Indoor fan target frequency
Code 21	----	N/A
Code 22	----	N/A
Code 23	----	N/A
Code 24	----	N/A
Code 25	----	N/A
Code 26	----	N/A
Code 27	----	N/A
Code 28	----	N/A
Code 29	----	N/A
Code 30	----	N/A

To exit Inquiry Mode:

Press and hold **On/Off** and **Fan** together   for 2 seconds.

INDOOR UNIT DIAGNOSTIC GUIDES

Table 12 — Indoor Unit Diagnostic Guides

OPERATION LAMP (TIMES)	TIMER LAMP	DISPLAY	ERROR INFORMATION
☆1	OFF	EH 00/EH 0A	Indoor unit EEPROM parameter error
☆2	OFF	EL 01	Indoor/outdoor unit communication error
☆3	OFF	EH 02	Zero-crossing signal detection error
☆4	OFF	EH 03	Indoor fan operating outside of the normal range
☆5	OFF	EC 51	Outdoor unit EEPROM parameter error
☆5	OFF	EC 52	T3 is in open circuit or has short circuited
☆5	OFF	EC 53	T4 is in open circuit or has short circuited
☆5	OFF	EC 54	TP is in open circuit or has short circuited
☆5	OFF	EC 56	T2B is in open circuit or has short circuited
☆6	OFF	EH 60	T1 is in open circuit or has short circuited
☆6	OFF	EH 61	T2 is in open circuit or has short circuited
☆12	OFF	EC 07	Outdoor fan operating outside of the normal range
☆9	OFF	EH 0b	Indoor PCB/Display board communication error
☆8	OFF	EL 0C	Refrigerant leakage detection
☆7	FLASH	PC 00	IPM malfunction or IGBT OSCP
☆2	FLASH	PC 01	Over voltage or over low voltage protection
☆3	FLASH	PC 02	Compressor or IPM high temp/pressure protection
☆5	FLASH	PC 04	Inverter compressor drive error
☆1	FLASH	PC 08	Current overload protection
☆6	FLASH	PC 40	Comm. error between outdoor chip and compressor chip
☆7	FLASH	PC 03	Low pressure protection
☆1	ON	--	Indoor units mode conflict
☆9	OFF	EH b1	Indoor board and Multi-function communication error
☆11	OFF	FH 0d	Ionizer malfunction

O (on – light) X (off – light) ☆ (flash)

DIAGNOSIS AND SOLUTION (CONT)

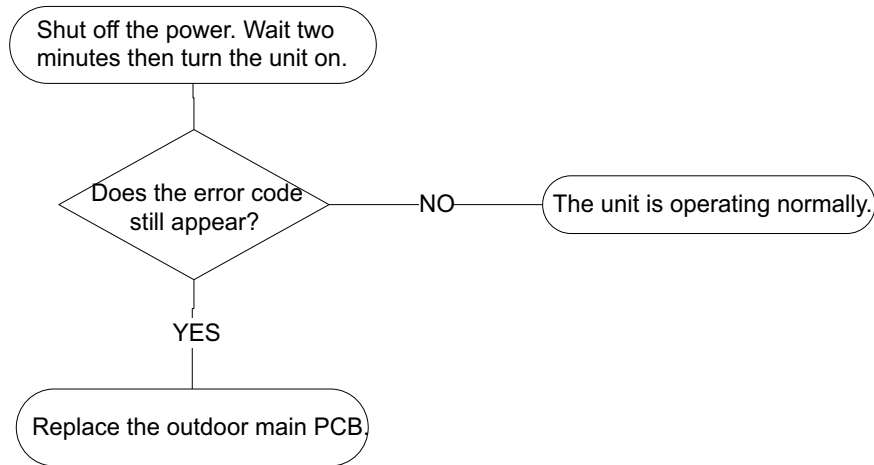
Outdoor EEPROM Parameter Error or Compressor Driven Chip EEPROM Parameter Error (EC51)

Description: Outdoor PCB main chip does not receive feedback from the EEPROM chip or the compressor driven chip.

Recommended parts to repair:

- Outdoor PCB

Troubleshooting



A221413

DIAGNOSIS AND SOLUTION (CONT)

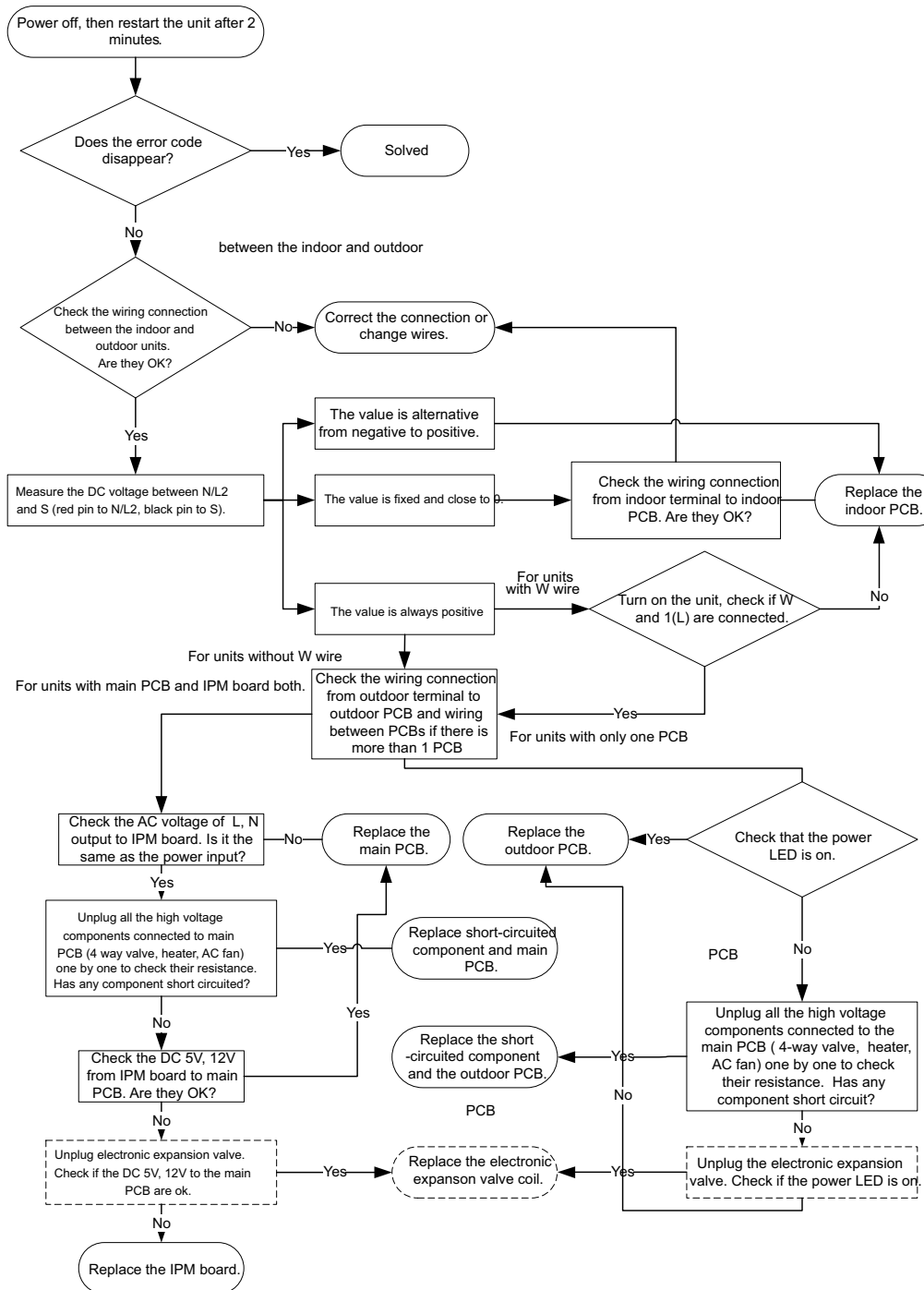
Indoor and Outdoor Unit Communication Error (EL01)

Description: The indoor unit cannot communicate with the outdoor unit

Recommended parts to repair:

- Indoor PCB
- Outdoor PCB
- Short-circuited component

Troubleshooting



A221414

For certain models, the outdoor PCB could not be removed separately. In this case, the outdoor electric control box should be replaced as a whole.

DIAGNOSIS AND SOLUTION (CONT)

Remarks:

- Use a multimeter to test the DC voltage between the 2 port (or S or L2 port) and 3 port (or N or S port) of the outdoor unit.
- The red pin of the multimeter connects with 2 port (or S or L2 port) while the black pin is for 3 port (or N or S port) the unit is running normal, the voltage is moving alternately as positive values and negative values.
- If the outdoor unit malfunctions, the voltage remains in a narrow positive value.
- If the indoor unit malfunctions, the voltage maintains a fixed value.

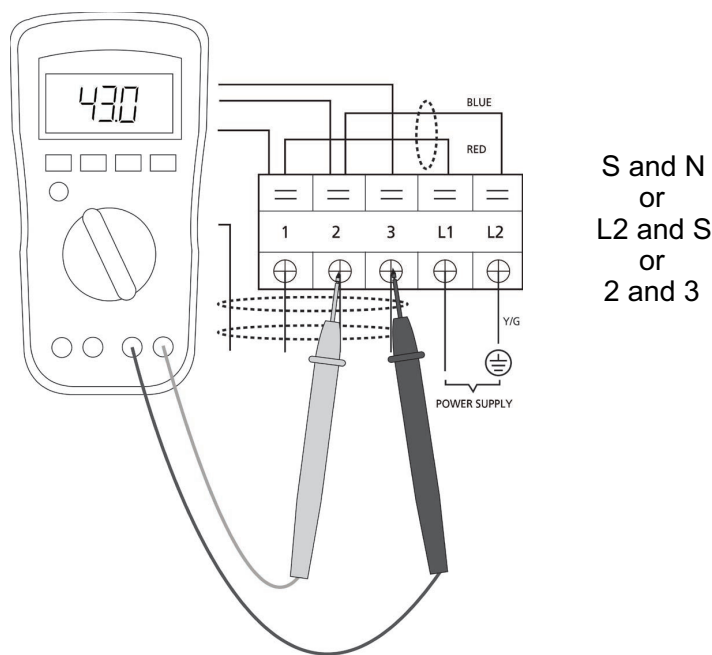


Fig. 19 — Multimeter

A221415

- Use a multimeter to test the reactor's resistance when disconnected from the capacitor.
- The normal value should be around zero ohm. Otherwise, the reactor must have a malfunction. Check the reactor and ensure it is not shorted to the ground.

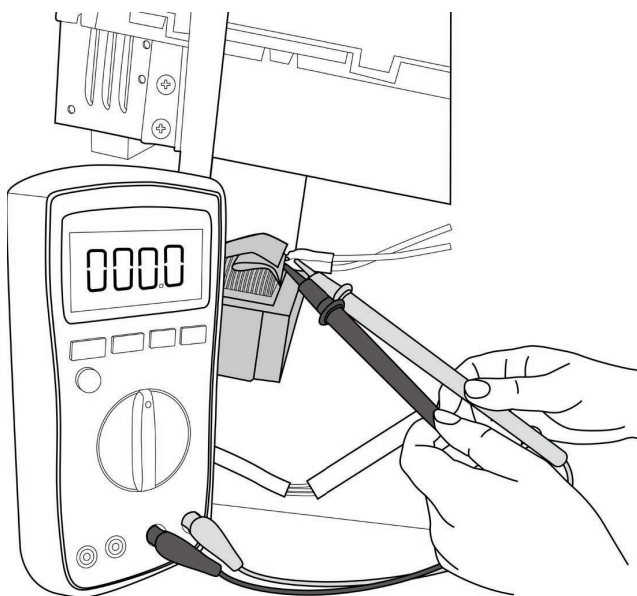


Fig. 20 – Multimeter

A221428

DIAGNOSIS AND SOLUTION (CONT)

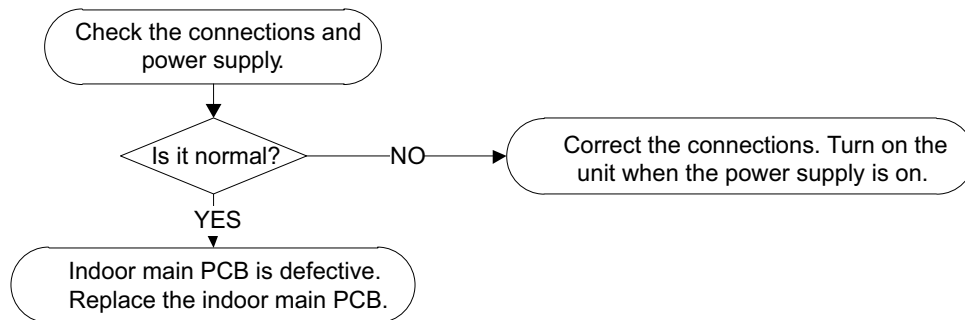
Zero Crossing Detection Error Diagnosis and Solution (EH02)

Description: When the PCB does not receive a zero crossing signal feedback for 4 minutes or the zero crossing signal time interval is abnormal.

Recommended parts to repair:

- Connection wires
- Indoor main PCB

Troubleshooting



A221429

NOTE: Zero crossing detection error is only valid for the unit with an AC fan motor. For other models, this error does not apply.

DIAGNOSIS AND SOLUTION (CONT)

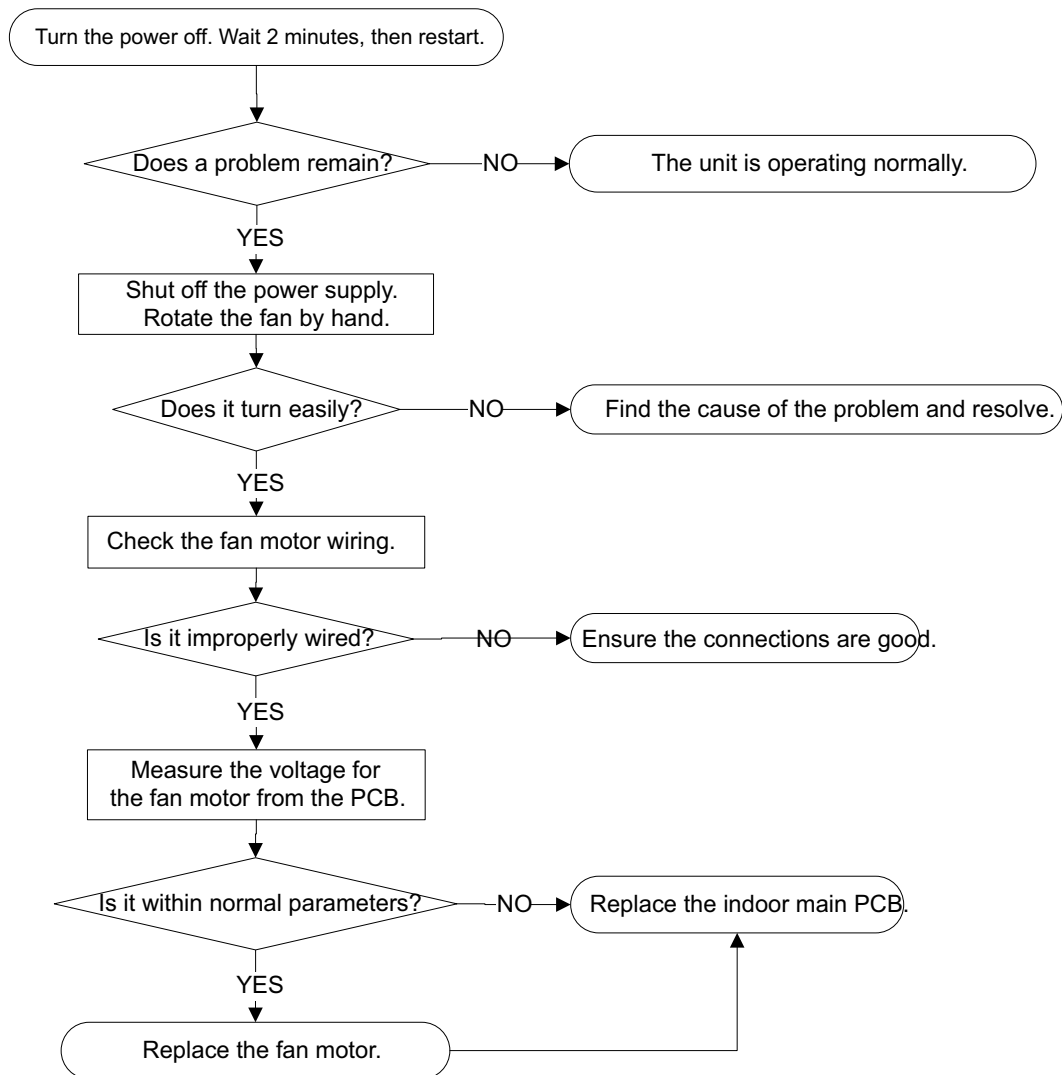
The indoor fan speed is operating outside of the normal range (EH03)

Description: When the indoor fan speed remains too slow or too fast for an extended period of time, the LED displays a failure code and the unit turns off.

Recommended parts to repair:

- Connection wires
- Indoor main PCB
- Fan assembly
- Indoor main PCB

Troubleshooting



A221430

DIAGNOSIS AND SOLUTION (CONT)

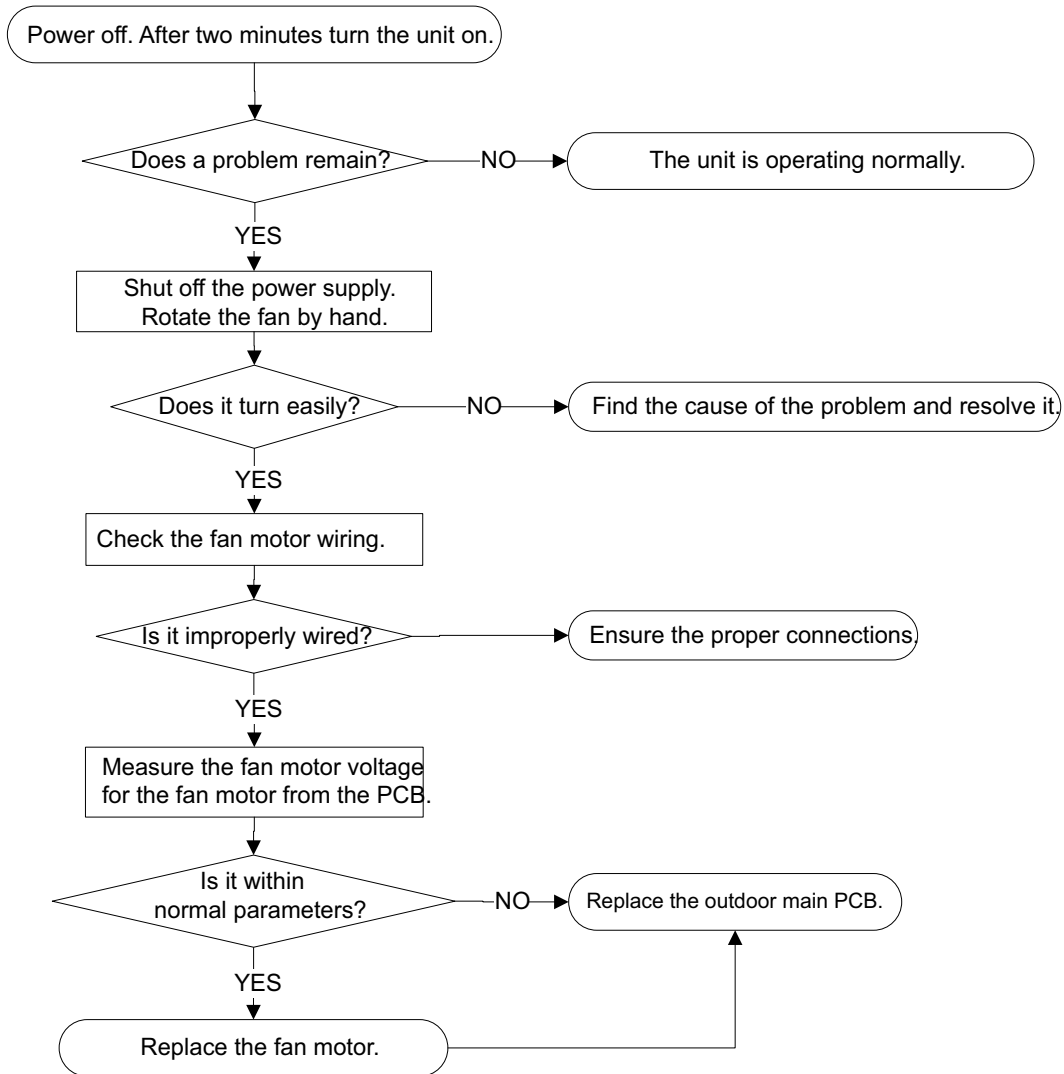
The Outdoor Fan Speed is Operating Outside of Normal Range (EC07)

Description: When the outdoor fan speed remains too low or too high for a certain time, the LED displays the failure code and the AC turns off.

Recommended parts to repair:

- Connection wires
- Fan assembly
- Fan motor
- Outdoor main PCB

Troubleshooting



A221431

NOTE: For certain models, the outdoor PCB can not be removed separately. In this case, the outdoor electric control box should be replaced as a whole.

DIAGNOSIS AND SOLUTION (CONT)

Outdoor DC Fan Motor (DC motor that controls the chip on the PCB)

- 1. Release the UVW connector. Measure the resistance of U-V, U-W, V-W. If the resistance is not equal to each other, the fan motor is faulty and must be replaced. Otherwise, proceed to step 2.
- 2. Power on the unit and when the unit is in **STANDBY** mode, measure the pin4-5 voltage in the feedback signal connector. If the value is not 5V, change the PCB. Otherwise, proceed to step 3.
- 3. Rotate the fan by hand, measure the pin1-5, pin 2-5 and pin 3-5 voltage levels in the feedback signal connector. If any voltage is not in the positive voltage fluctuation, the fan motor is faulty and must be replaced.

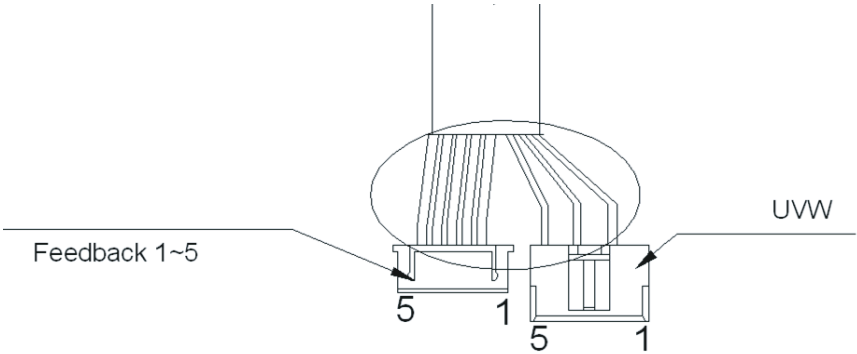


Fig. 21 — Outdoor DC Fan Motor (DC motor that controls the chip on the PCB)

A221432

NO.	1	2	3	4	5
Color	Orange	Grey	White	Pink	Black
Signal	Hu	Hv	Hw	Vcc	GND

Color	Red	Blue	Yellow
Signal	W	V	U

DIAGNOSIS AND SOLUTION (CONT)

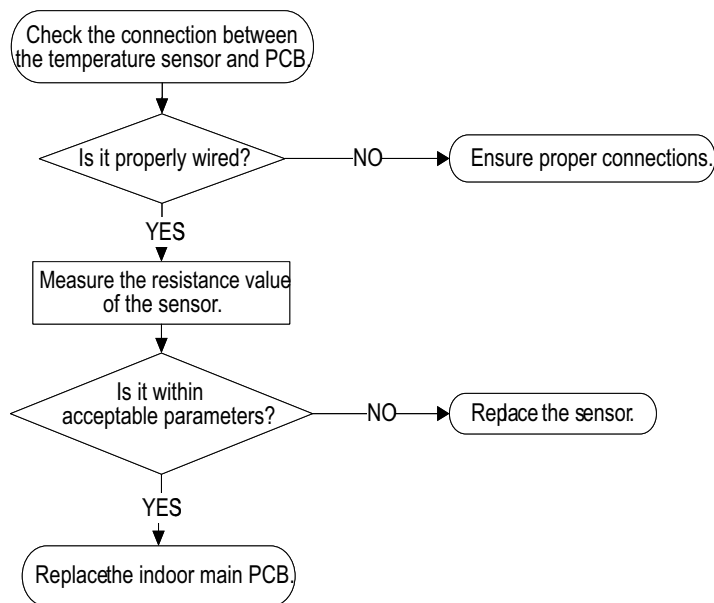
Indoor Temperature Sensor Is an Open Circuit or a Short Circuit (T1, T2) (EH60)

Description: If the sampling voltage is lower than 0.06V or higher than 4.94V, the LED displays the failure code.

Recommended parts to repair:

- Connection wires
- Sensors
- Indoor main PCB

Troubleshooting



A221433

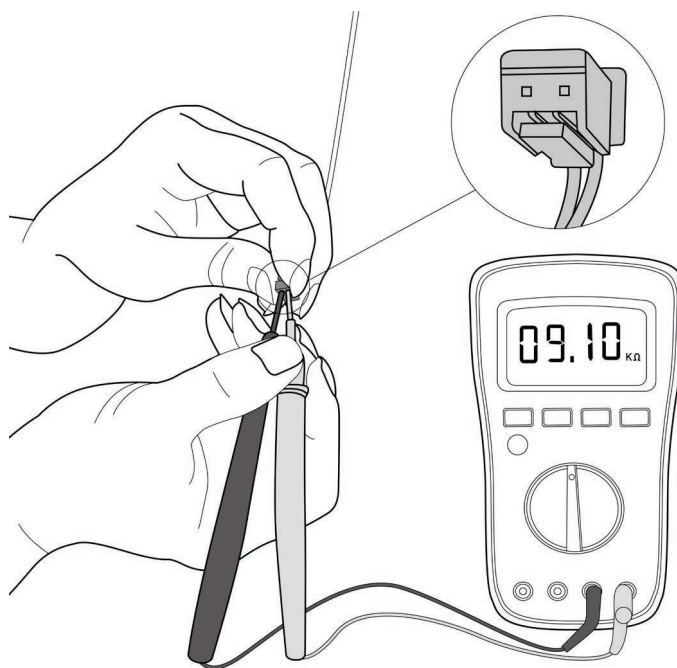


Fig. 22 — Test

A221434

NOTE: Figure 22 and the value shown within are for reference only. This note applies to all images and their values within this document.

DIAGNOSIS AND SOLUTION (CONT)

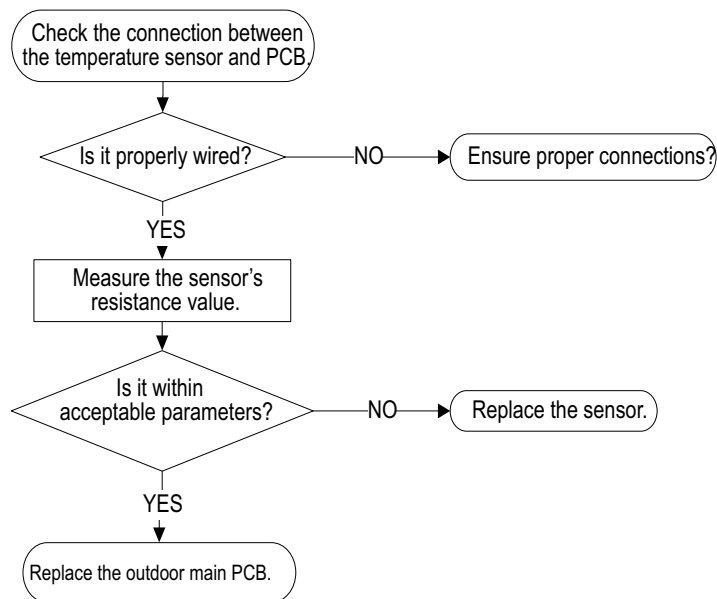
Outdoor Temperature Sensor Is an Open Circuit or Short Circuited (T3, T4, TP, T2B, TH) (EC53)

Description: If the sampling voltage is lower than 0.06V or higher than 4.94V, the LED displays the failure code.

Recommended parts to repair:

- Connection wires
- Sensors
- outdoor main PCB

Troubleshooting



A221435

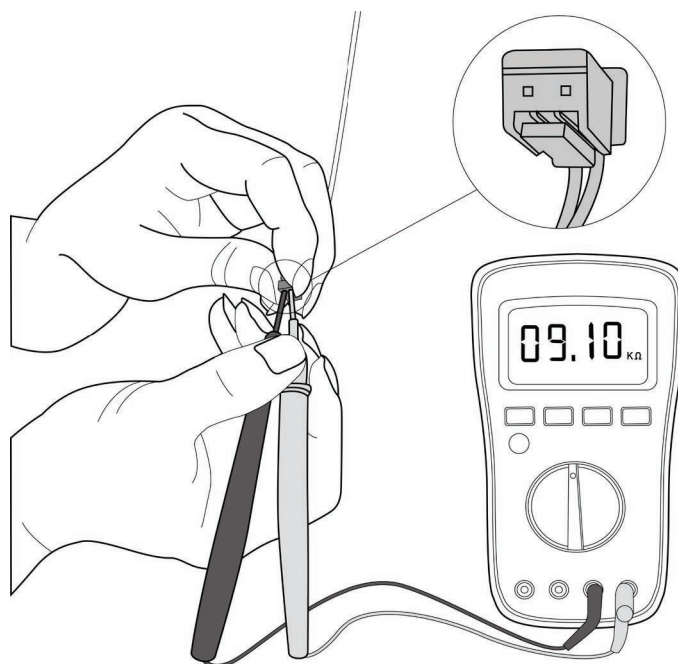


Fig. 23 —Test

A221436

NOTE: For certain models, the outdoor PCB could not be removed separately. In this case, the outdoor electric control box should be replaced as a whole. For certain models, the outdoor unit uses a combination sensor, T3,T4 and TP are the same sensor. Figure 23 and the value are for reference only.

DIAGNOSIS AND SOLUTION (CONT)

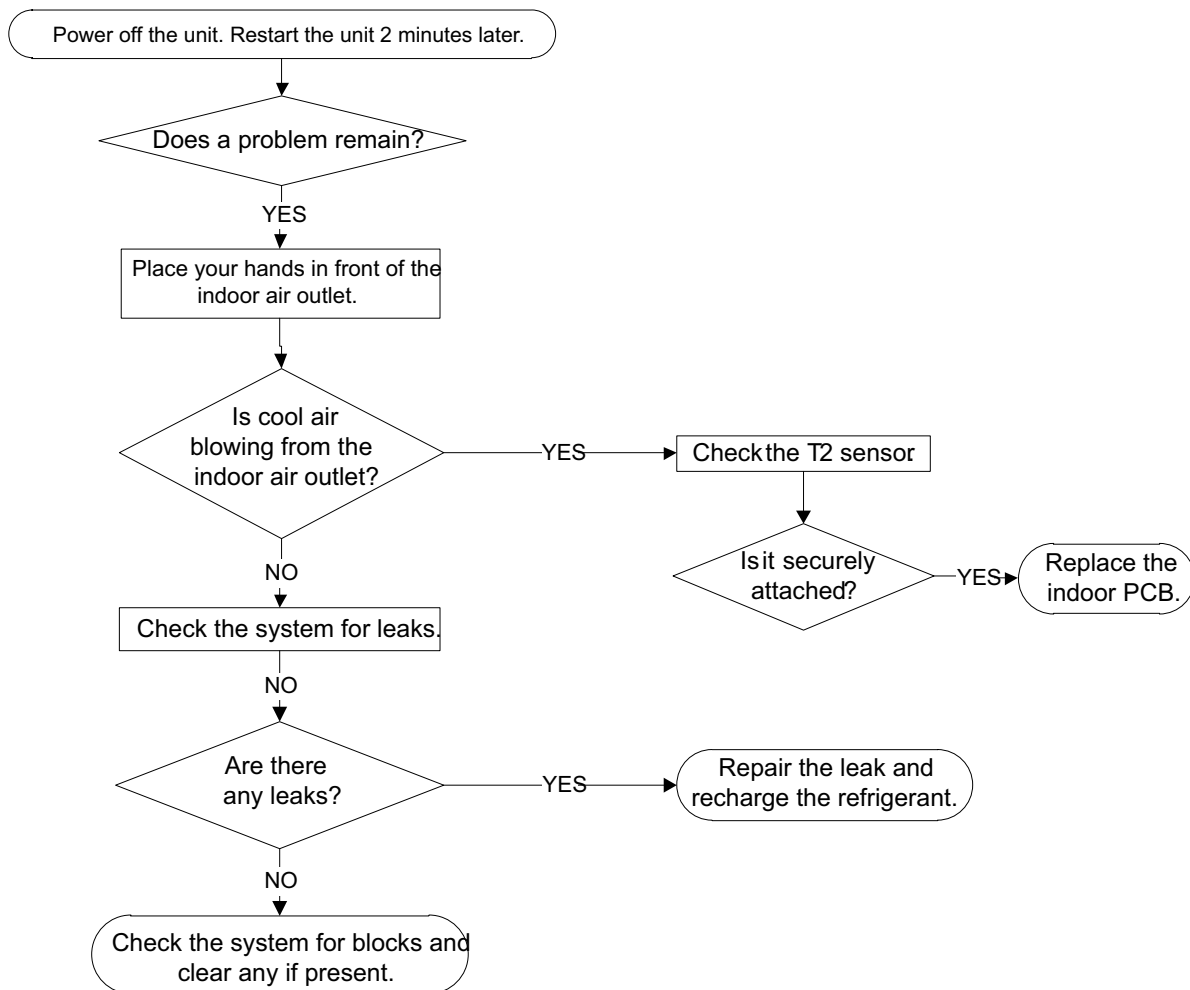
Refrigerant Leakage Detection (EL0C)

Description: Define the compressor's evaporator coil temperature (T2) starts running as Tcool. In the initial 5 minutes after the compressor starts, if $T2 < T_{cool} - 1.8^{\circ}\text{F}$ (1°C) is not maintained for 4 seconds and the compressor runs at a frequency higher than 50Hz however it does not maintain for a minimum of three minutes and this issue occurs 3 times, the LED displays the failure code and the unit turns off.

Recommended parts to repair:

- T2 Sensor
- Indoor PCB
- Additional refrigerant

Troubleshooting



A221437

DIAGNOSIS AND SOLUTION (CONT)

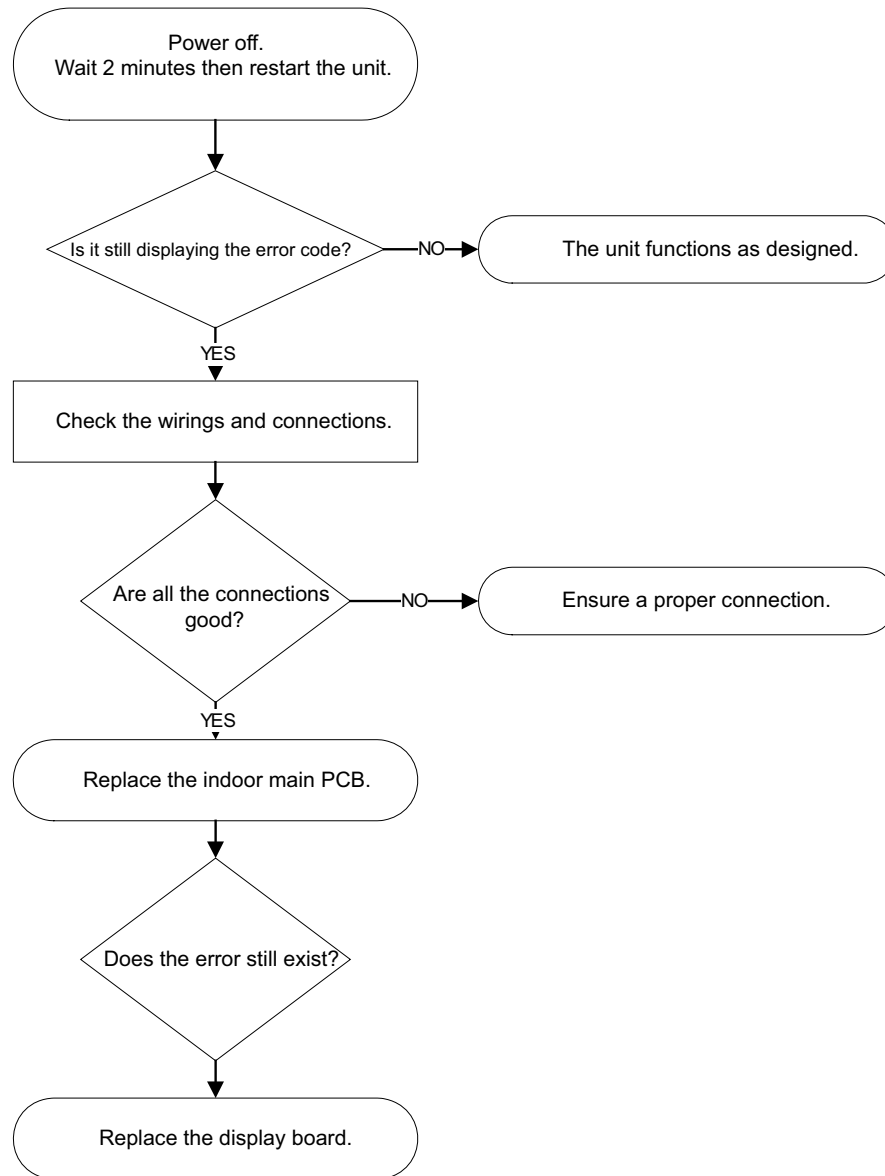
Indoor PCB/Display Board Communication Error (EH06)

Description: The indoor PCB does not receive feedback from the display board.

Recommended parts to repair:

- Communication wire
- Indoor PCB
- Display board

Troubleshooting



A221438

DIAGNOSIS AND SOLUTION (CONT)

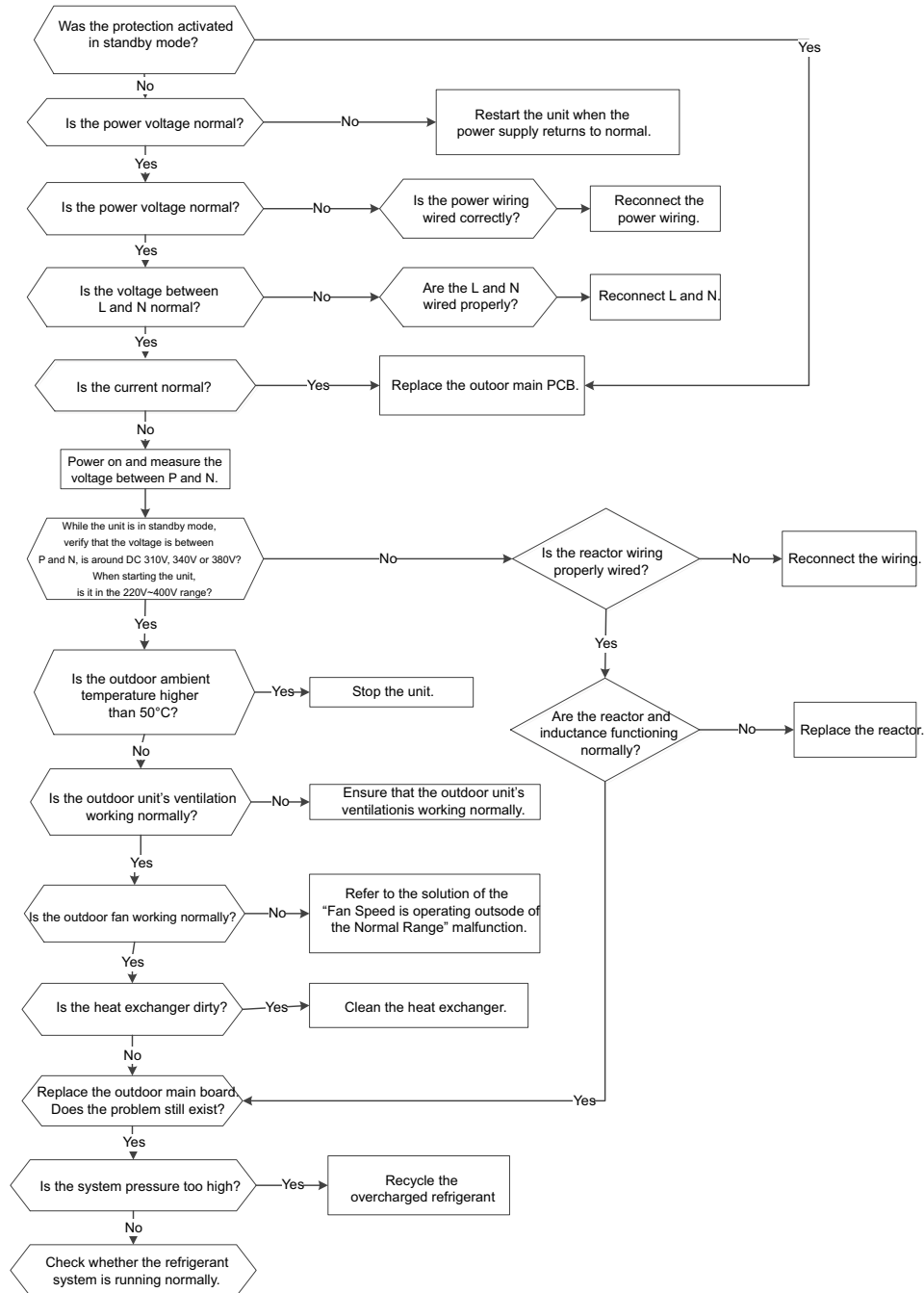
Current Overload Protection (PC08)

Description: An abnormal current rise is detected by checking the specified detection circuit.

Recommended parts to repair:

- Communication wires
- Reactor
- Outdoor fan
- Outdoor PCB

Troubleshooting



A221439

NOTE: For certain models, the outdoor PCB can not be removed separately. In this case, the outdoor electric control box should be replaced as a whole.

DIAGNOSIS AND SOLUTION (CONT)

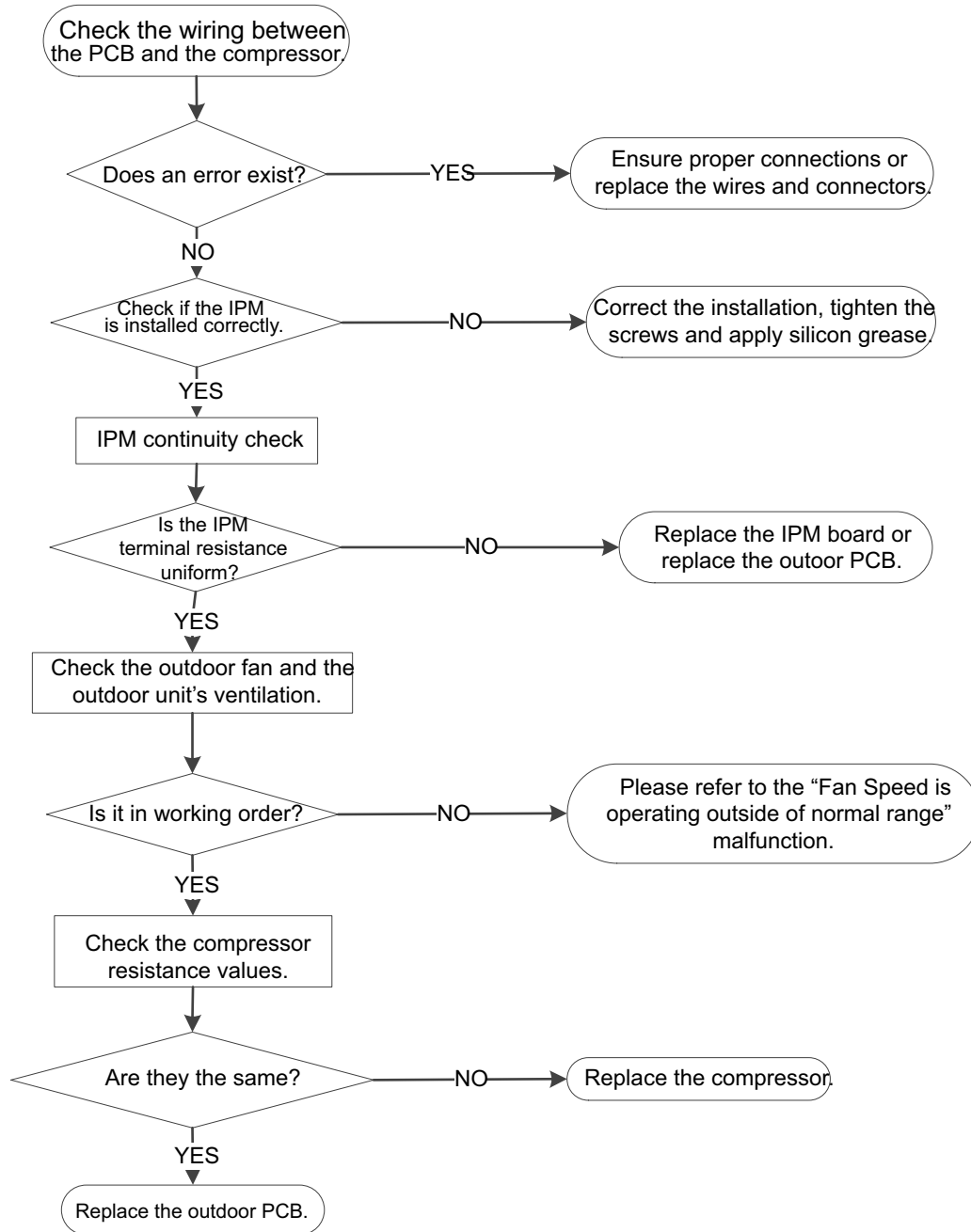
IPM Malfunction or IGBT over-strong current protection (PC00)

Description: If the IPM sends an abnormal voltage signal to the compressor drive chip, the LED displays the failure code and the unit turns off.

Recommended parts to repair:

- Communication wires
- IPM module board
- Outdoor fan assembly
- Compressor
- Outdoor PCB

Troubleshooting



A221440

DIAGNOSIS AND SOLUTION (CONT)

Index

IPM Continuity Check



WARNING

Electricity remains in the capacitors even when the power supply is off.
Ensure the capacitors are fully discharged before troubleshooting.

1.

Turn off the outdoor unit and disconnect the power supply.
2.

Discharge the electrolytic capacitors and ensure all the energy storage has been discharged.
3.

Disassemble the outdoor PCB or disassemble the IPM board.
4.

Measure the resistance value between P and U (V,W,N), U (V,W) and N.

DIGITAL TESTER		RESISTANCE VALUE	DIGITAL TESTER		RESISTANCE VALUE
(+) Red	(-) Black	∞ (Several MW)	(+) Red	(-) Black	∞ (Several MW)
P	N		U	N	
	U		V		
	V		W		
	W		-		

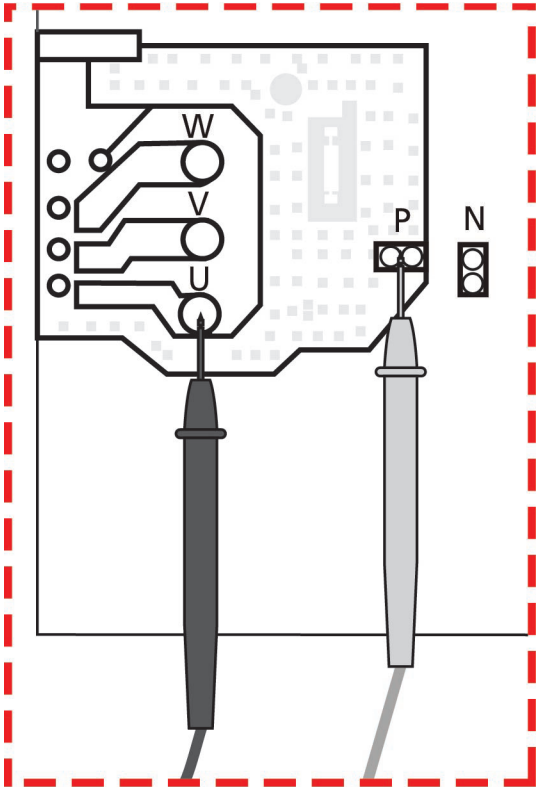


Fig. 24 — N-W

A221441

DIAGNOSIS AND SOLUTION (CONT)

Compressor Check

Disconnect the compressor and check the resistance between U-V, V-W and U-W, and all 3 values should be equal. If not, the compressor is faulty and needs to be replaced.

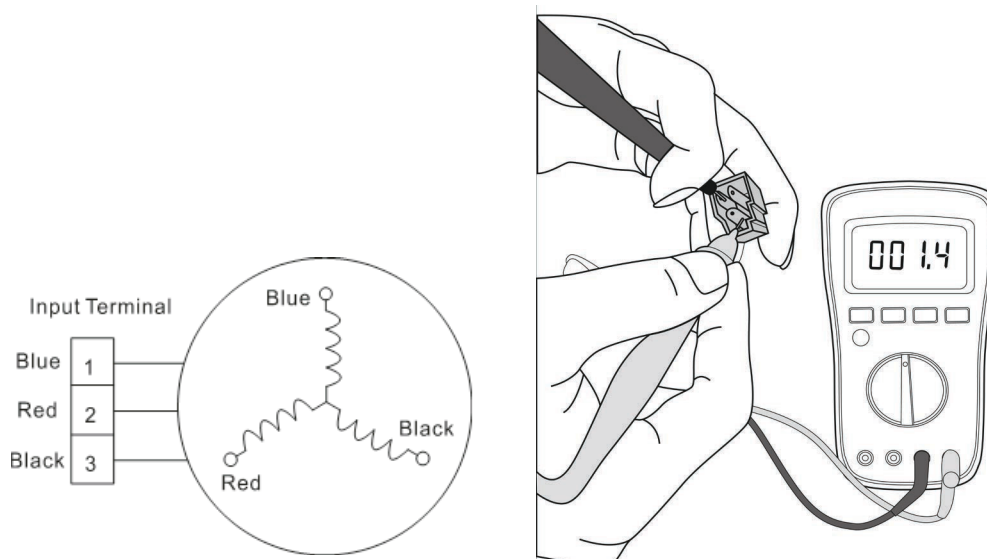


Fig. 25 — Compressor Checks

NOTE: Figure 25 is for reference only.

DIAGNOSIS AND SOLUTION (CONT)

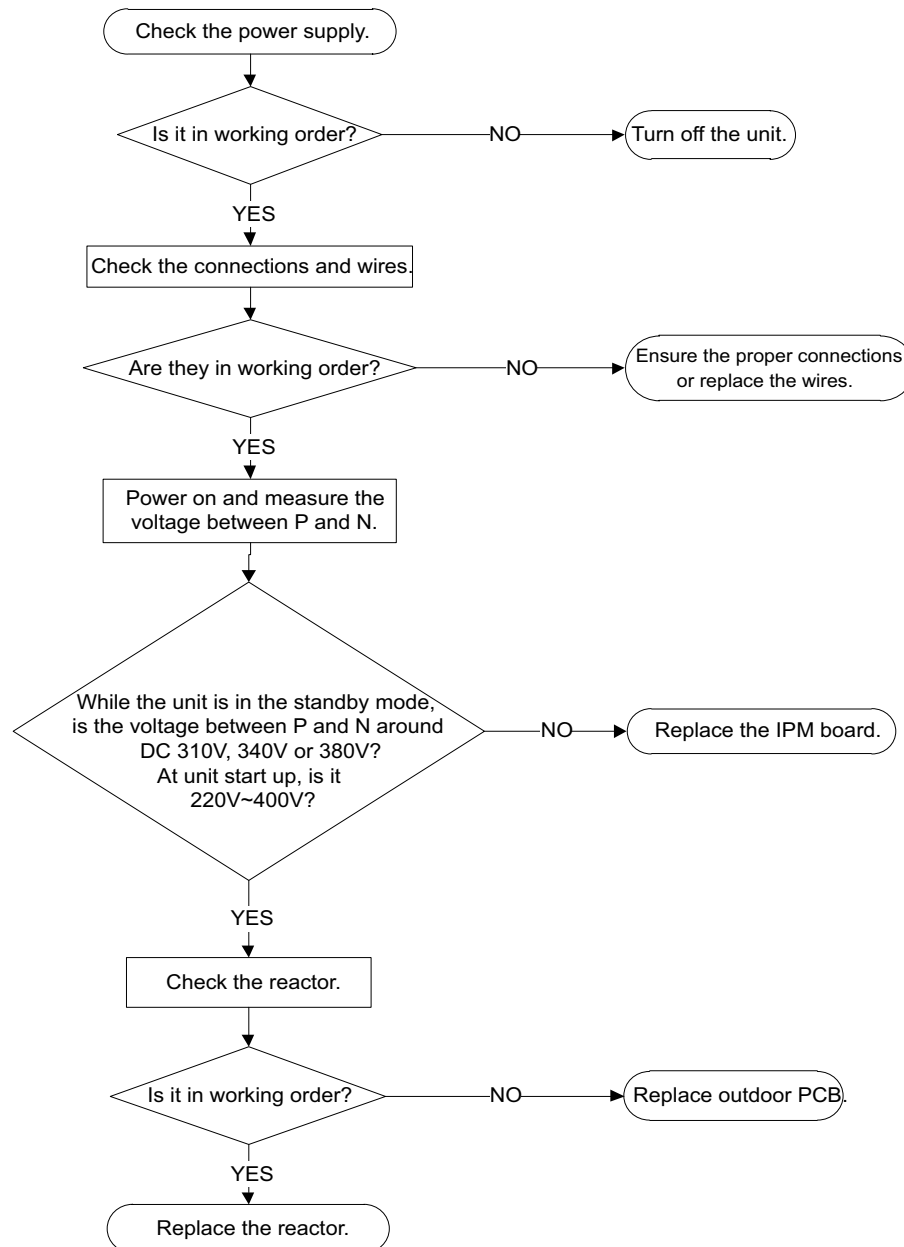
Over Voltage or Low Voltage Protection (PC01)

Description: Abnormal increases or decreases in voltage are detected by checking the specified voltage detection circuit.

Recommended parts to repair:

- Power supply wires
- IPM module board
- PCB
- Reactor

Troubleshooting



A221443

DIAGNOSIS AND SOLUTION (CONT)

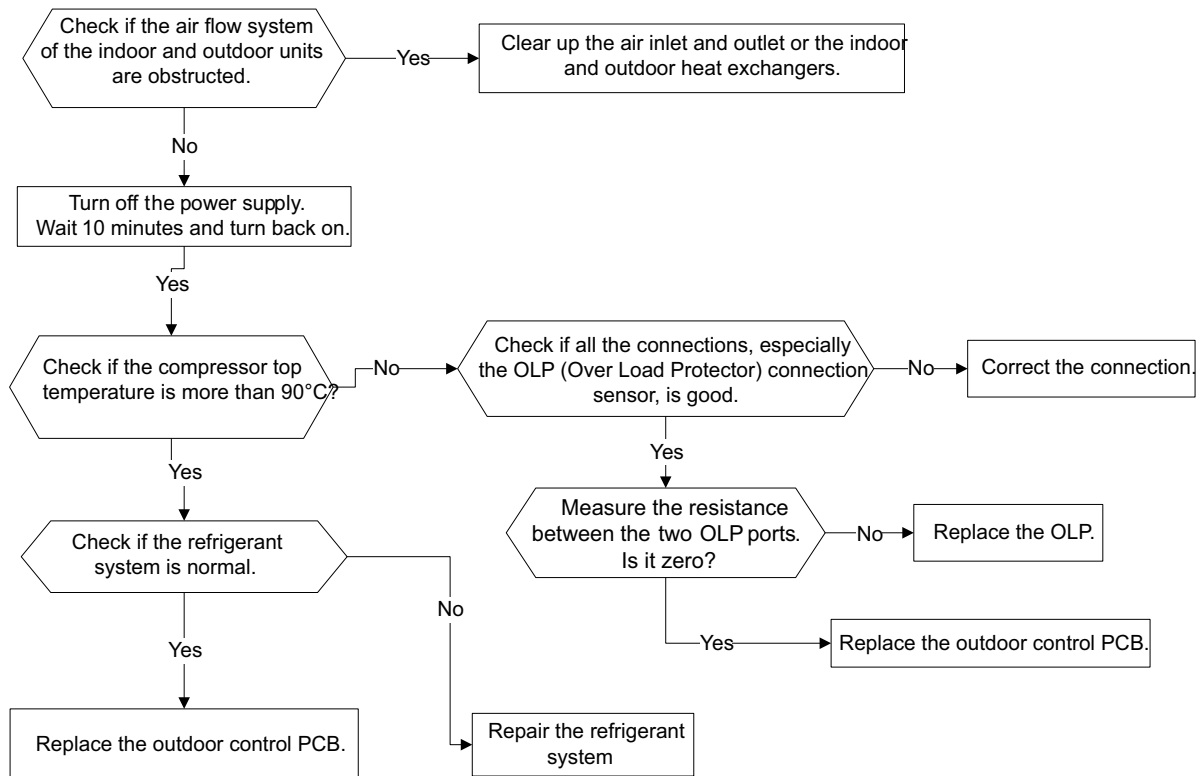
Top Temperature Protection for Compressor or High Temperature or High Pressure Protection of IPM Module (PC02)

Description: For some models with overload protection, if the sampling voltage is not 5V, the LED displays the failure. If the IPM module temperature is higher than a certain value, the LED displays the failure code. For some models with a high pressure switch, the outdoor pressure switch cuts off the system when the pressure is higher than 4.4 MPa and the LED displays the failure code.

Recommended parts to repair:

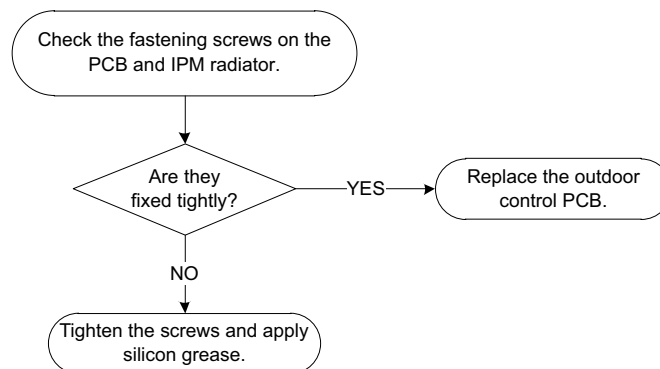
- Connection wires
- Outdoor PCB
- IPM module board
- High pressure protector
- System blockages

Troubleshooting



A221444

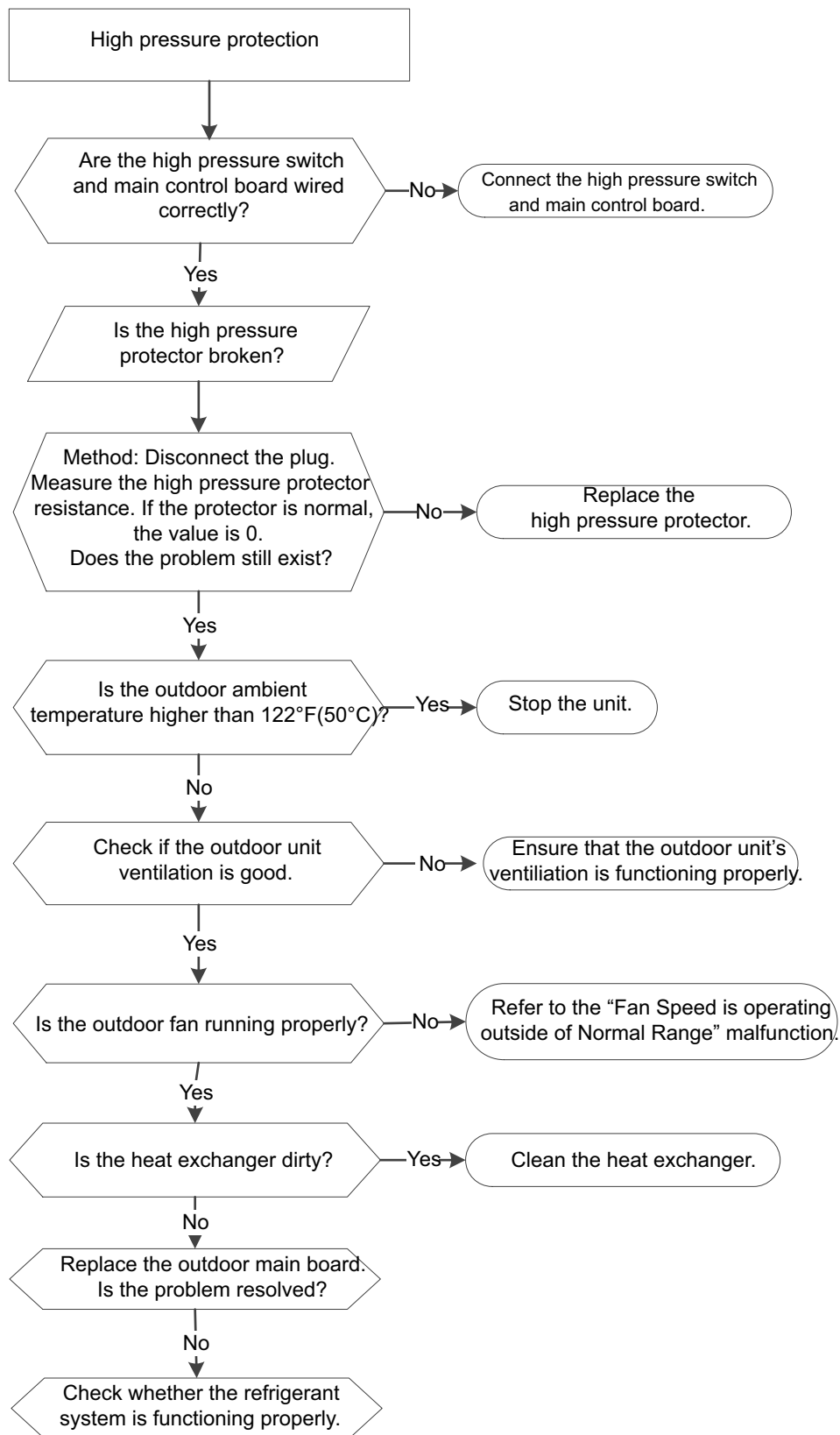
NOTE: For certain models, the outdoor PCB cannot be removed separately. In this instance, the outdoor electric control box should be replaced as a whole.



A221445

DIAGNOSIS AND SOLUTION (CONT)

Top temperature protection for compressor or High temperature or High pressure protection of IPM module (PC02) (cont.)



A221446

DIAGNOSIS AND SOLUTION (CONT)

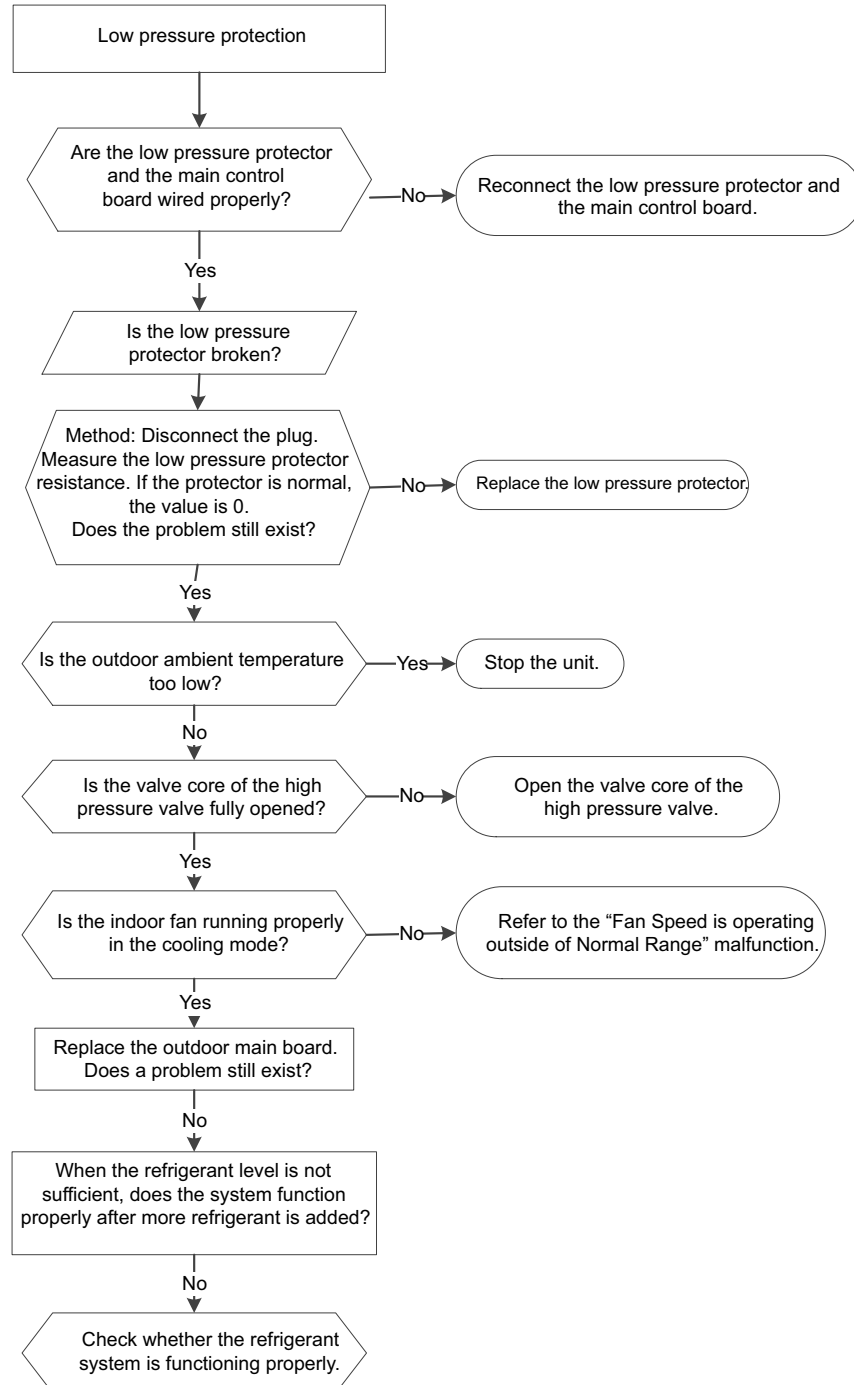
Low Pressure Protection (PC03)

Description: The outdoor pressure switch shuts the unit down because the low pressure is lower than 0.13 MPa and the LED displays the failure code.

Recommended parts to repair:

- Connection wires
- Outdoor PCB
- Low pressure protector
- Refrigerant

Troubleshooting



DIAGNOSIS AND SOLUTION (CONT)

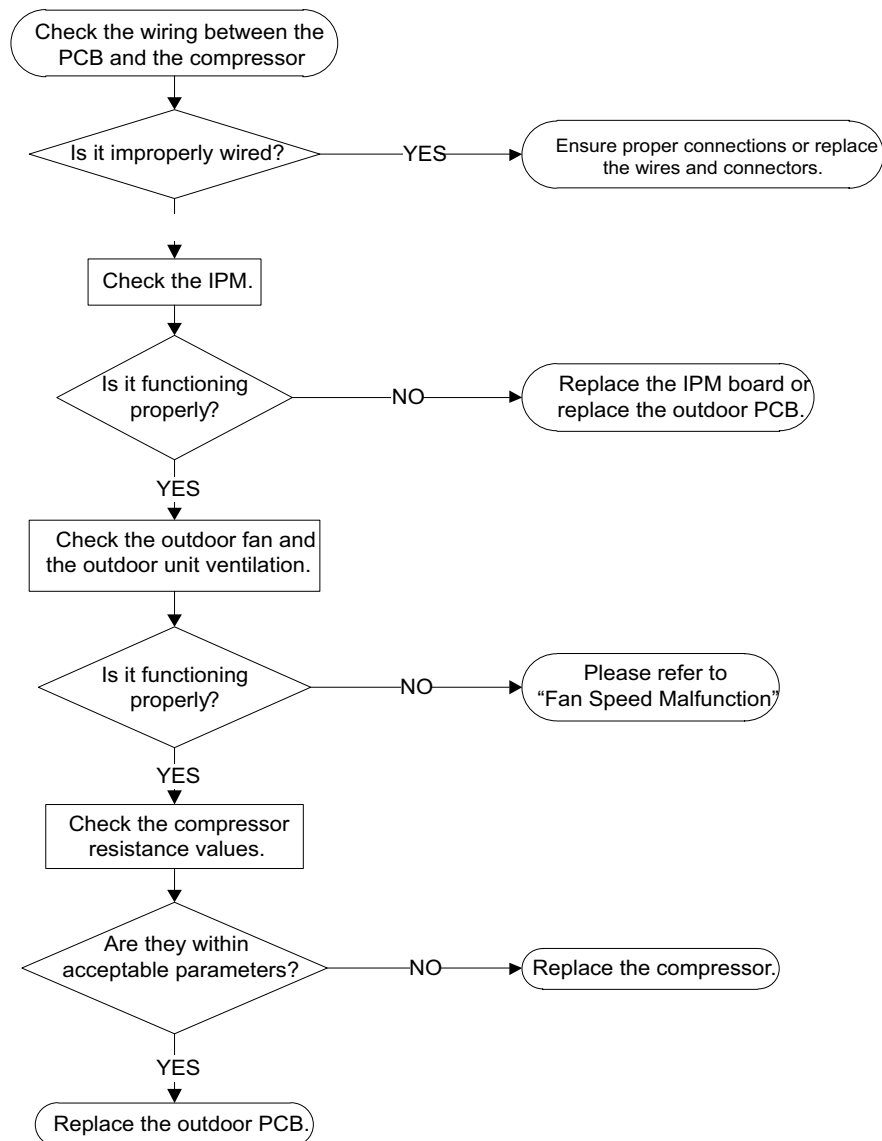
Inverter compressor Drive Error (PC04)

Description: An abnormal inverter compressor drive is detected by a special detection circuit, including communication signal detection, voltage detection, compressor rotation and speed signal detection.

Recommended parts to repair:

- Connection wires
- IPM module board
- Outdoor fan assembly
- Compressor
- Outdoor PCB

Troubleshooting

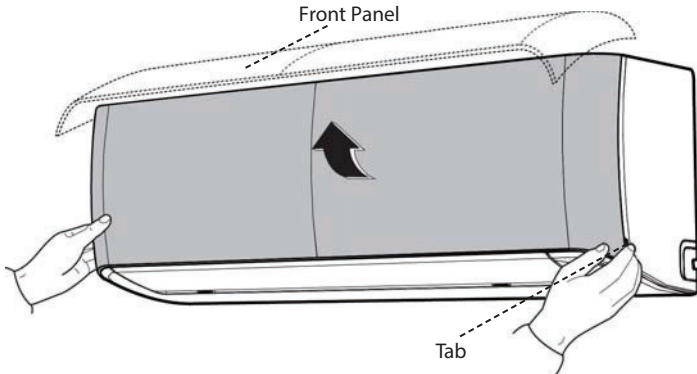
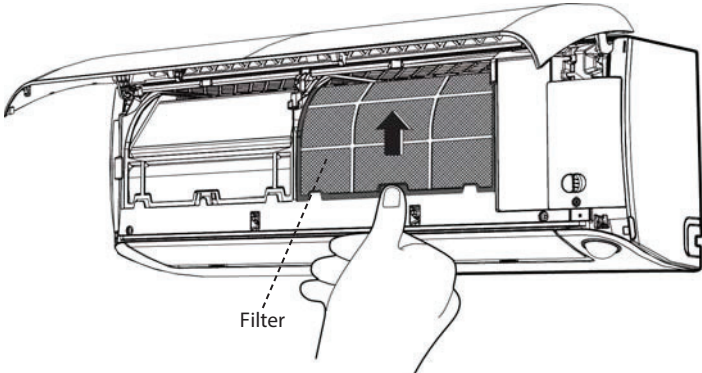


A221448

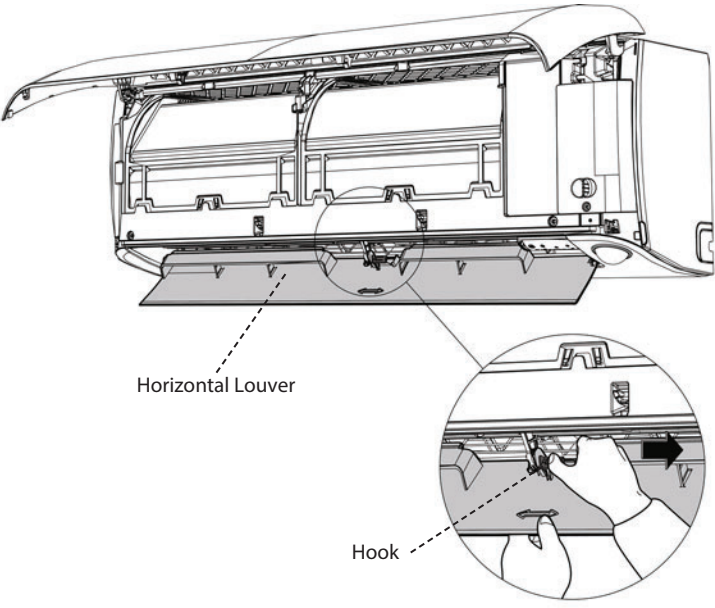
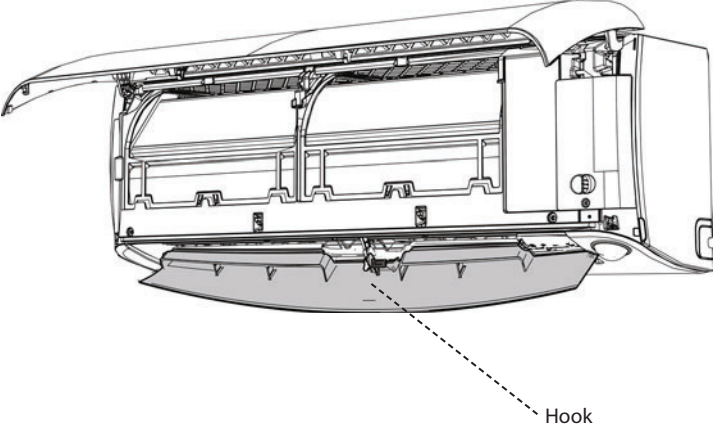
DISASSEMBLY INSTRUCTIONS

Indoor Unit Sizes 09K-12K

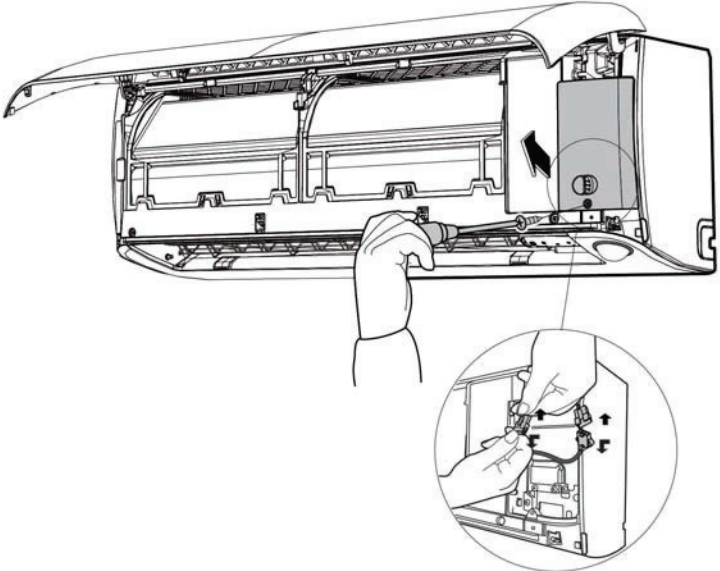
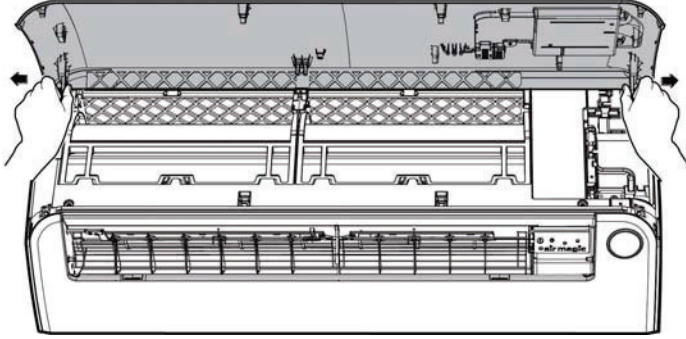
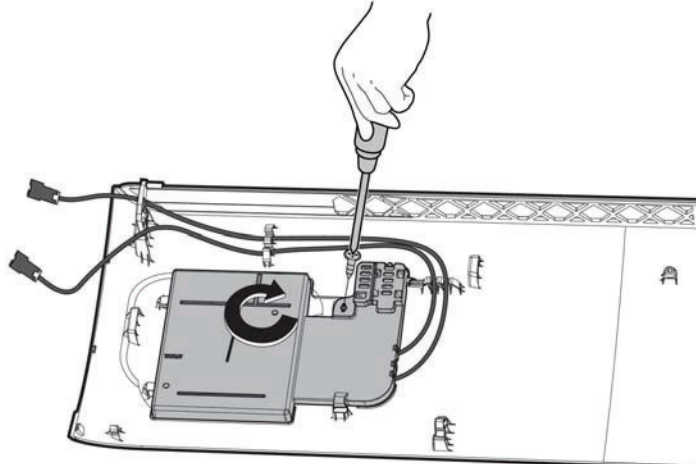
(Front Panel)

Procedure	Illustration
1) Hold the front panel by the tabs on both sides and lift it up.	 The illustration shows a hand holding the left side of the front panel, which is being lifted upwards. A dashed line points to the top edge of the panel, labeled 'Front Panel'. Another dashed line points to a small tab on the bottom right corner, labeled 'Tab'. An upward-pointing arrow is shown on the front panel.
2) Push up the bottom of the air filter while pressing the bottom of the unit, and then pull the air filter out (down).	 The illustration shows the front panel removed, revealing the internal air filter. A hand is shown pushing up the bottom of the filter. A dashed line points to the filter, labeled 'Filter'. An upward-pointing arrow is shown on the filter.

DISASSEMBLY INSTRUCTIONS (CONT)**Indoor Unit Sizes 09K-12K (continued)****Front Panel**

Procedure	Illustration
3) Open the horizontal louver and push the hook towards the right to open.	 The illustration shows a side view of the indoor unit's front panel with the horizontal louver partially open. A dashed line points to the louver, labeled 'Horizontal Louver'. A circular inset shows a close-up of a hand pushing a hook to the right, with an arrow indicating the direction of movement. A dashed line points to the hook, labeled 'Hook'.
4) Bend the horizontal louver lightly with both hands to loosen the hooks, then remove the horizontal louver.	 The illustration shows the front panel with the horizontal louver bent downwards. A dashed line points to the hook, labeled 'Hook'.

DISASSEMBLY INSTRUCTIONS (CONT)**Indoor Unit Sizes 09K-12K (continued)****Front Panel**

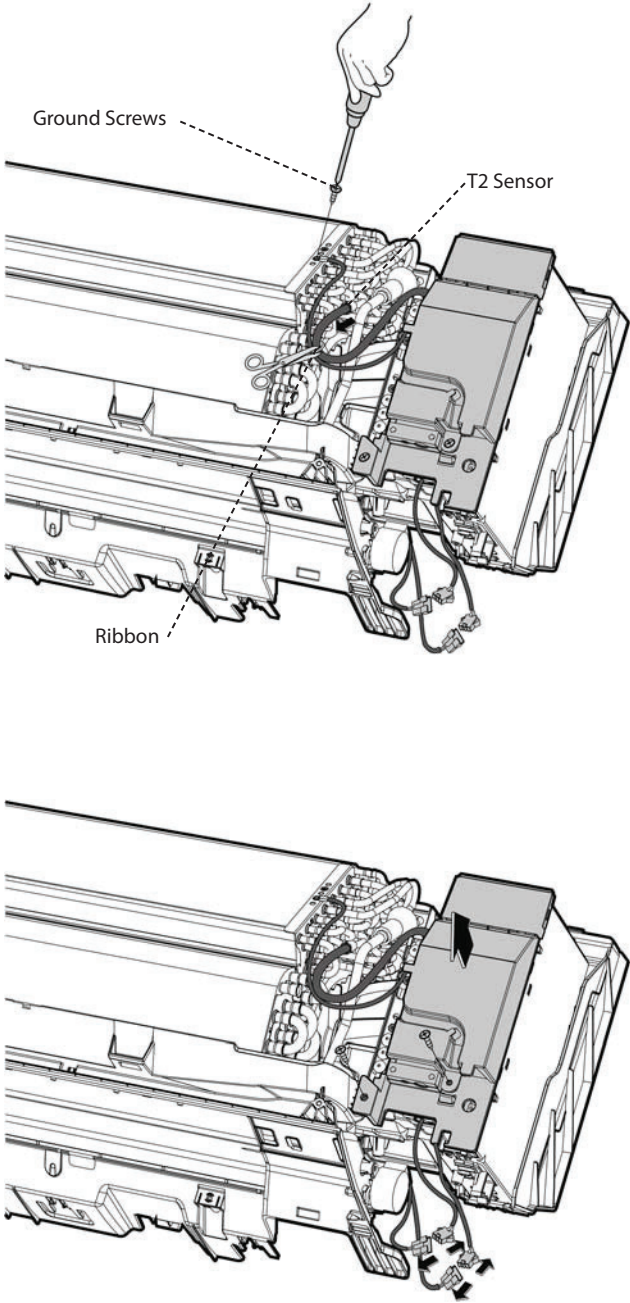
Procedure	Illustration
5) Remove the screw (1) then remove the electrical cover. 6) Disconnect the two connectors on the display board.	 The illustration shows the top of the indoor unit with the front panel partially removed. A hand is shown using a screwdriver to remove a screw from the electrical cover. An inset circular diagram shows a close-up of the display board with two connectors being disconnected.
7) Slide the front panel side to side to release each axis.	 The illustration shows the front panel being slid horizontally. Arrows indicate the direction of movement. A large upward-pointing arrow is positioned above the panel, indicating the next step in the disassembly process.
8) Remove the screw (1) and rotate the display board clockwise. Next, remove the clips.	 The illustration shows the display board being rotated clockwise. A hand is shown using a screwdriver to remove a screw from the board. A circular arrow indicates the direction of rotation.

DISASSEMBLY INSTRUCTIONS (CONT)

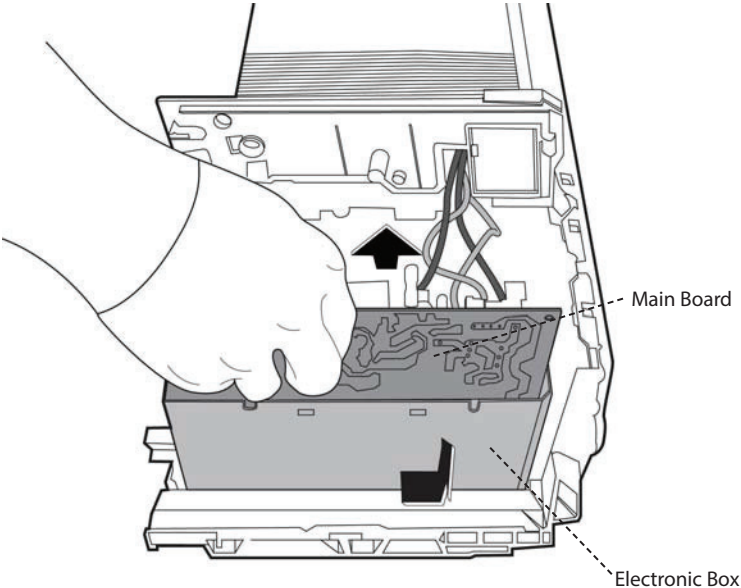
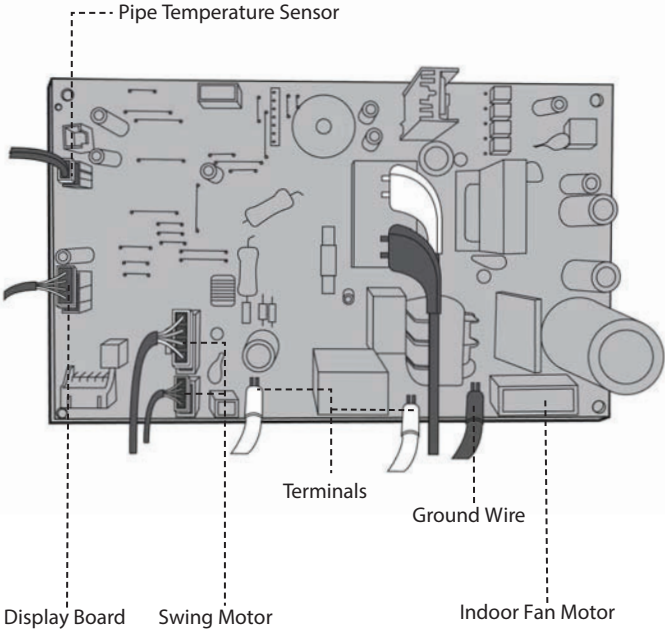
Indoor Unit Sizes 09K-12K (continued) (Anti-static gloves must be worn)

Electrical Parts

NOTE: Remove the front panel (refer to the front panel removal steps) before disassembling electrical parts.

Procedure	Illustration
1) Use a shear to cut the ribbon, then pull out the coil temperature sensor (T2). 2) Remove the screws (2) used for the ground connection.	

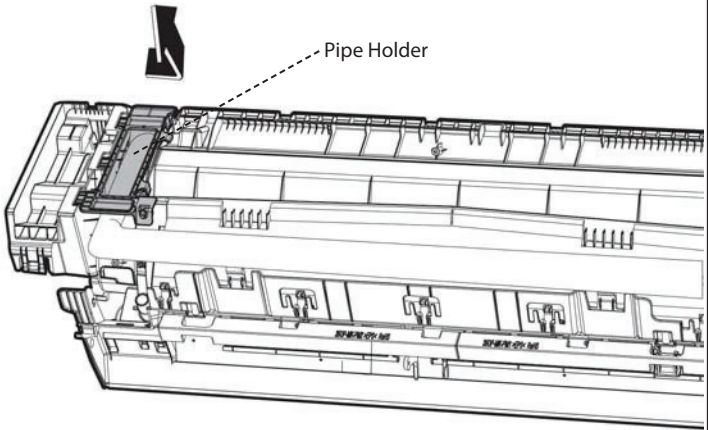
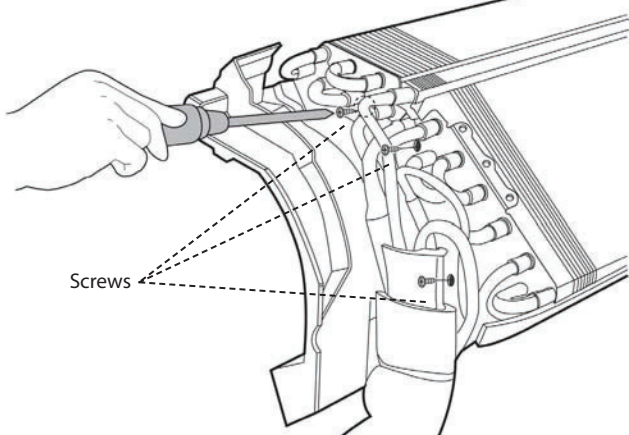
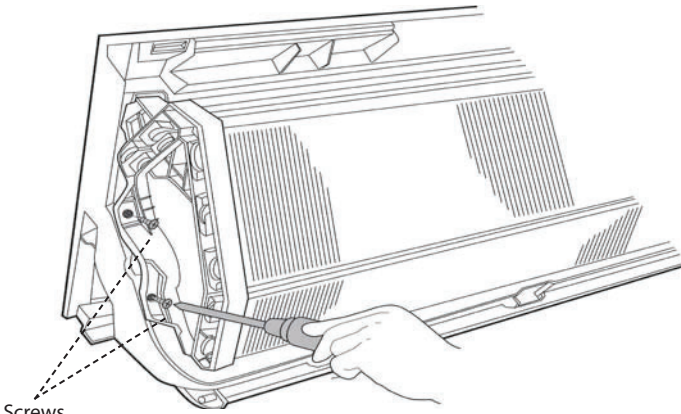
DISASSEMBLY INSTRUCTIONS (CONT)
Indoor Unit Sizes 09K-12K (continued) (Anti-static gloves must be worn)
Electrical Parts

Procedure	Illustration
5) Disconnect the wires. Next, remove the electronic main box. 6) Pull out the electrical box along the direction, indicated in the adjacent image, to remove it.	 Main Board Electronic Box
7) Each port connector is indicated in the adjacent image.	 Pipe Temperature Sensor Display Board Swing Motor Terminals Ground Wire Indoor Fan Motor

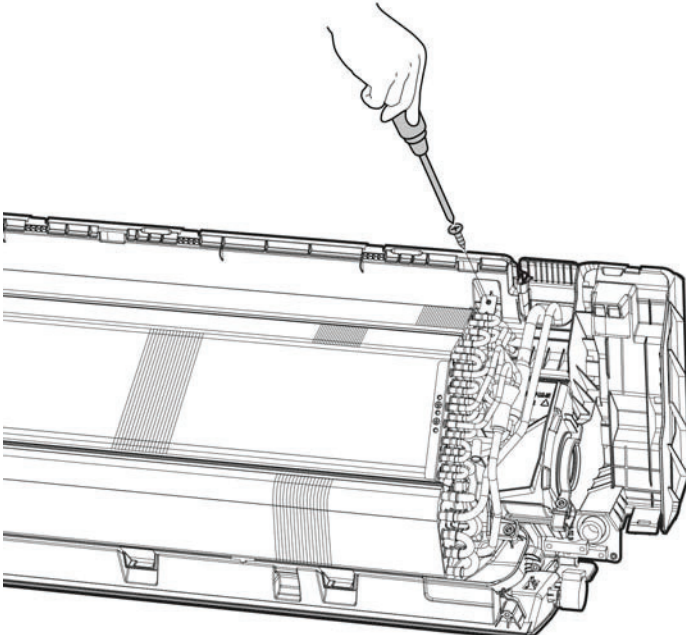
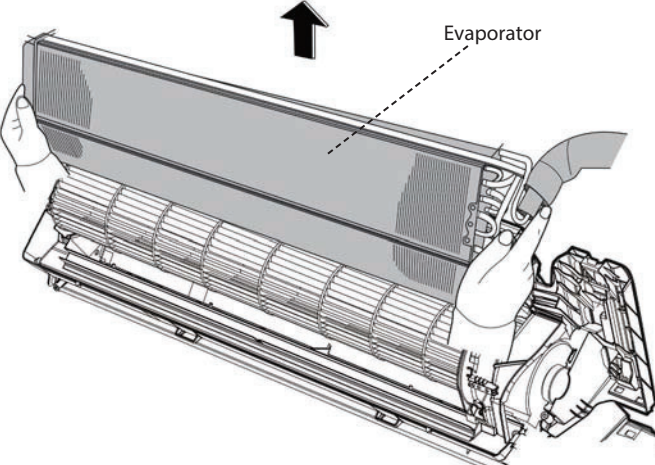
DISASSEMBLY INSTRUCTIONS (CONT)

Indoor Unit Sizes 09K-12K (continued)

Evaporator

Procedure	Illustration
1) Disassemble the pipe holder located at the rear of the unit.	 Pipe Holder 2) Remove the screws (3) on the evaporator located on the fixed plate.  Screws 3) Remove the screws (2) on the evaporator located at the base of the bearing side.  Screws

DISASSEMBLY INSTRUCTIONS (CONT)**Indoor Unit Sizes 09K-12K (continued)****Evaporator**

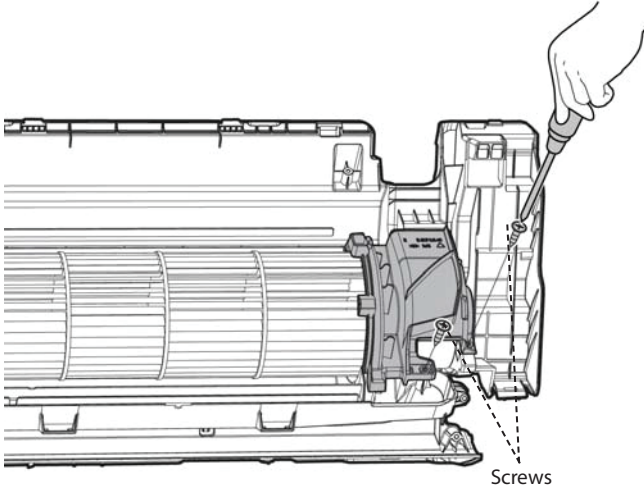
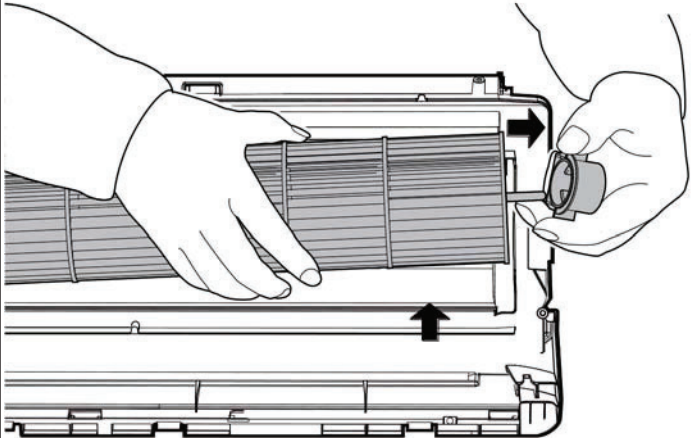
Procedure	Illustration
4) Remove the screw (1) on the evaporator located at the top of the evaporator.	
5) Remove the evaporator.	

DISASSEMBLY INSTRUCTIONS (CONT)

Indoor Unit Sizes 09K-12K

Fan Motor and Fan

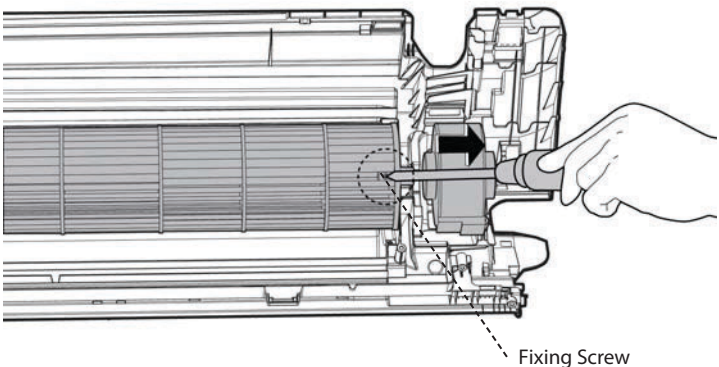
NOTE: Remove the front panel, electrical parts and evaporator (refer to front panel, electrical parts, and evaporator removal steps) before disassembling fan motor and fan.

Procedure	Illustration
1) Remove the screws (2) and remove the fan motor board.	 A line drawing of the indoor unit's internal components. A hand is using a screwdriver to remove two screws from a small circuit board (the fan motor board) located near the fan. Dashed lines point to the two screws being removed, with the label 'Screws' at the bottom right.
2) Remove the bearing sleeve.	 A line drawing showing a hand pulling the fan motor out of the unit. Another hand is shown pulling a small, cylindrical bearing sleeve off the motor's shaft. Arrows indicate the direction of removal for both the motor and the sleeve.

DISASSEMBLY INSTRUCTIONS (CONT)

Indoor Unit Sizes 09K-12K (continued)

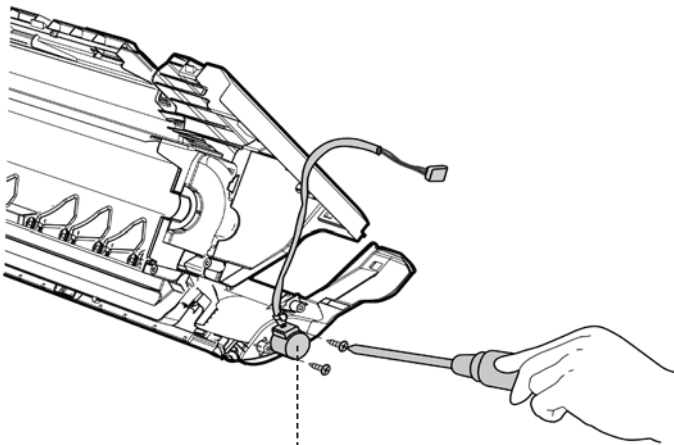
Fan Motor and Fan

Procedure	Illustration
3) Remove the set screw. 4) Pull out the fan motor and fan assembly from the side.	 Fixing Screw

Indoor Unit Sizes 09K-12K (continued)

Step Motor

NOTE: Remove the front panel and electrical parts (refer to the front panel, electrical parts removal steps) before disassembling the step motor.

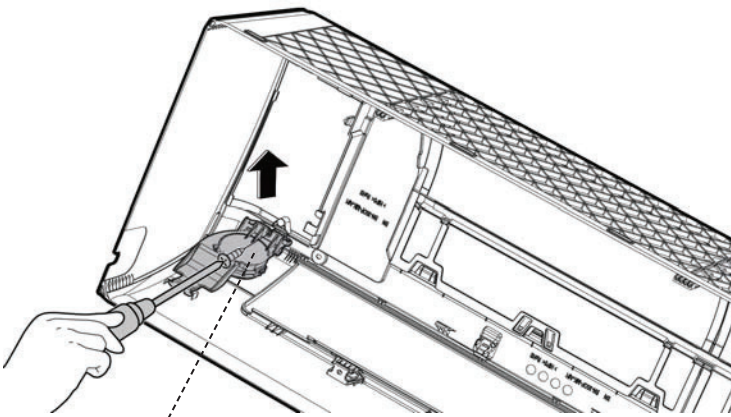
Procedure	Illustration
1) Remove the screws (2), then remove the stepping motor.	 Stepping Motor

DISASSEMBLY INSTRUCTIONS (CONT)

Indoor Unit Sizes 09K-12K (continued)

Occupancy Sensor

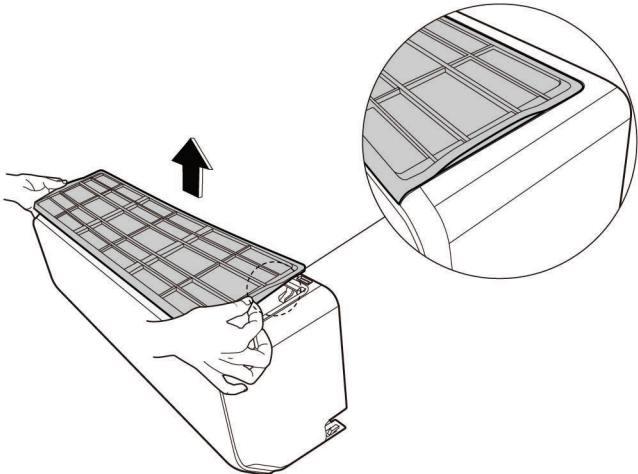
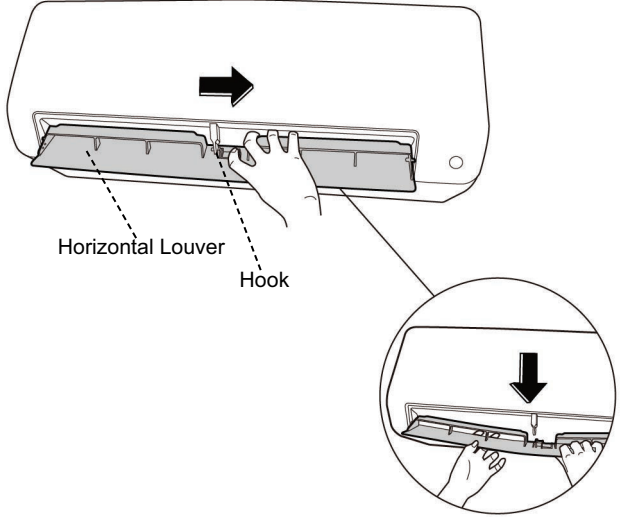
NOTE: Remove the front panel and electrical parts (refer to the front panel, electrical parts, evaporator and fan motor and fan removal steps) before disassembling the occupancy sensor.

Procedure	Illustration
1) Remove the screw (1), then remove the Occupancy Sensor.	 Occupancy Sensor

DISASSEMBLY INSTRUCTIONS (CONT)

Indoor Unit Size 18K

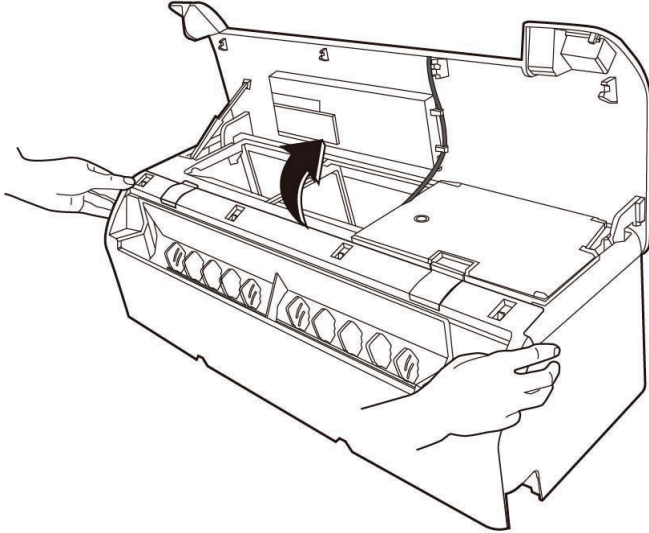
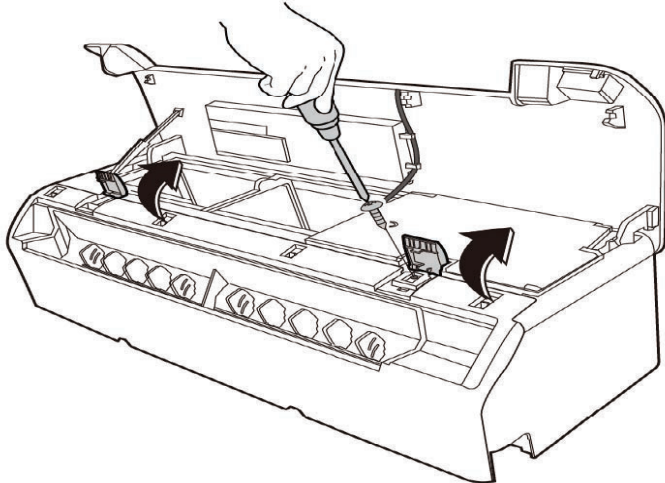
Front Panel

Procedure	Illustration
1) Place both hands along the filter's sides then pull the filter gently along the vertical direction, then remove it.	
2) Open the horizontal louver and push the locker towards the right to open. 3) Bend the horizontal louver slightly to loosen the hooks, then remove the horizontal louver.	

DISASSEMBLY INSTRUCTIONS (CONT)

Indoor Unit Size 18K (continued)

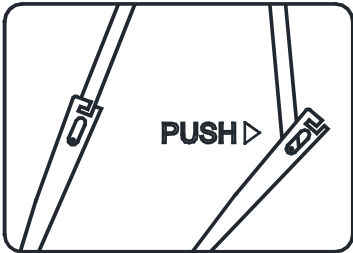
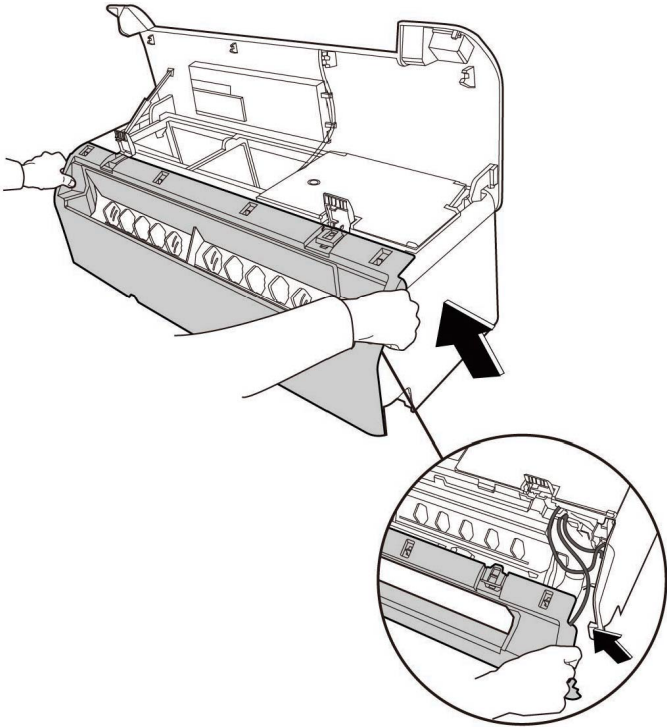
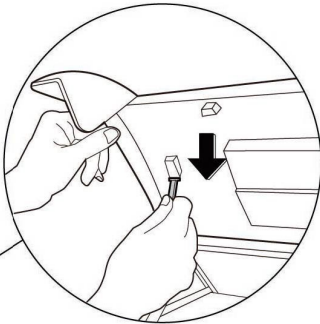
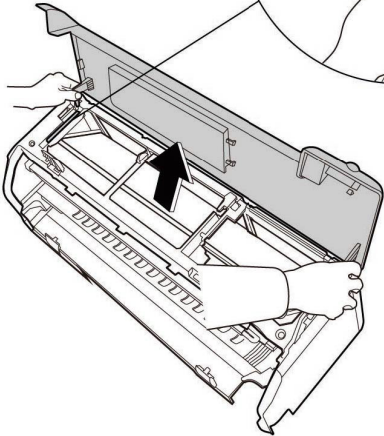
Front Panel

Procedure	Illustration
4) Open the panel assembly, move the slider to secure the panel.	
5) Open the two stop blocks of the panel frame assembly. 6) Remove 1 screw from the panel frame.	

DISASSEMBLY INSTRUCTIONS (CONT)

Indoor Unit Size 18K (continued)

Front Panel

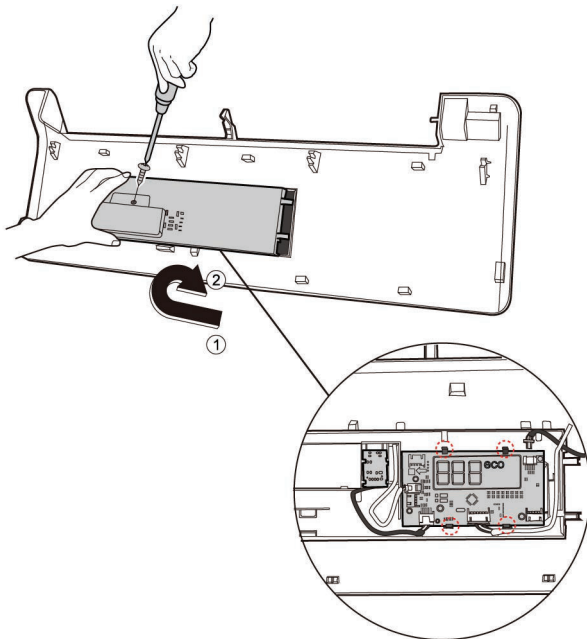
Procedure	Illustration
7) Pull the two sides of the bottom panel along the direction shown in the adjacent image to remove it. 8) Pull the panel's support bar to remove it. 9) Remove the panel assembly. Caution: To close the panel, bend the middle of the support bar, otherwise it will break. For 6K~18K models the support bar is located on the left side of the unit. For 24K and up models, the bar is located in middle of the unit. 	  

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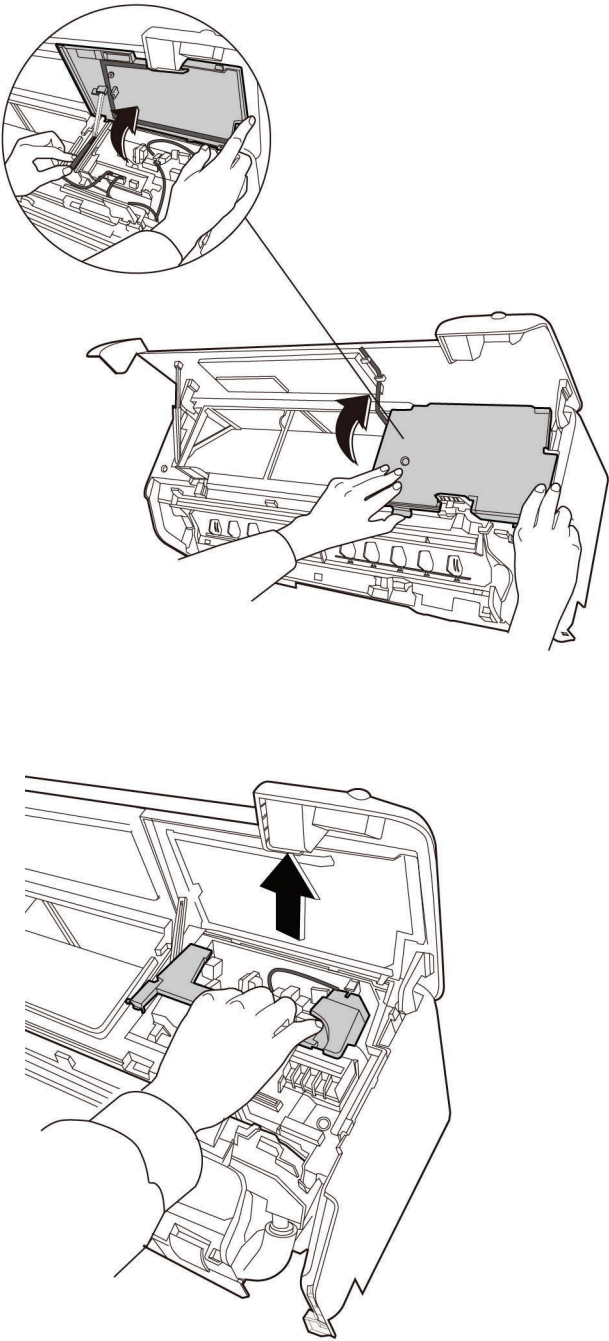
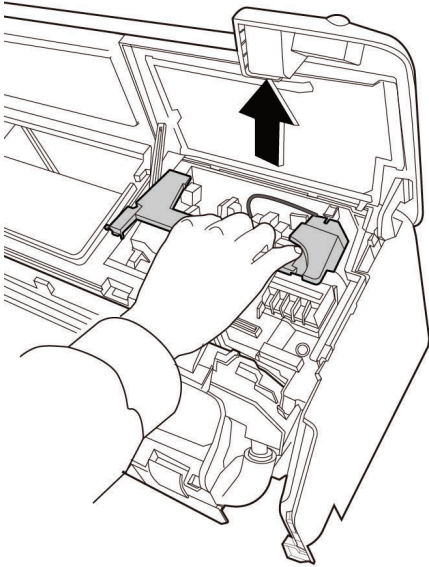
DISASSEMBLY INSTRUCTIONS (CONT)

Indoor Unit Size 18K (continued)

Front Panel

Procedure	Illustration
10) Remove 1 screw from the display board. 11) Rotate the display board subassembly in the direction shown in the adjacent picture. 12) Pull the four clips to remove the display board.	

DISASSEMBLY INSTRUCTIONS (CONT)**Electrical parts (Anti-static gloves must be worn)****Front Panel**

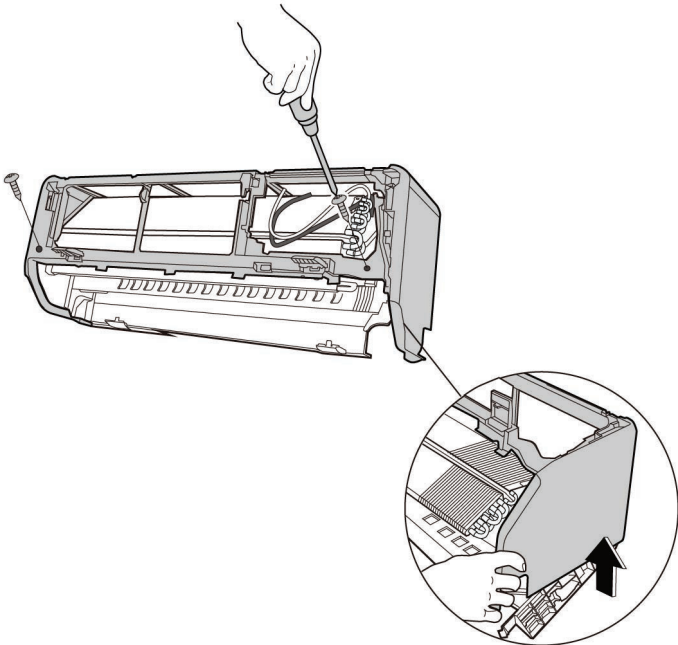
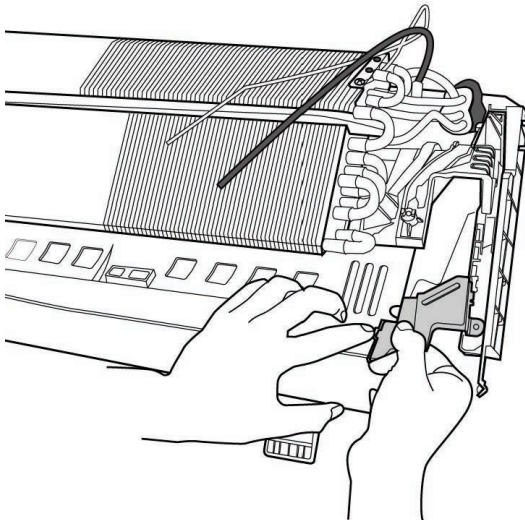
Procedure	Illustration
1) Pull the two ends of the electronic control box cover with your thumbs to open. 2) Raise the support bar to secure the cover.	
3) Pull the electrical control box holder to remove it.	

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DISASSEMBLY INSTRUCTIONS (CONT)

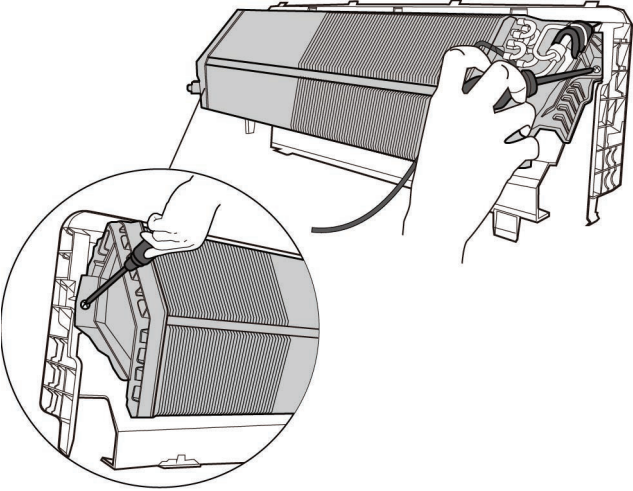
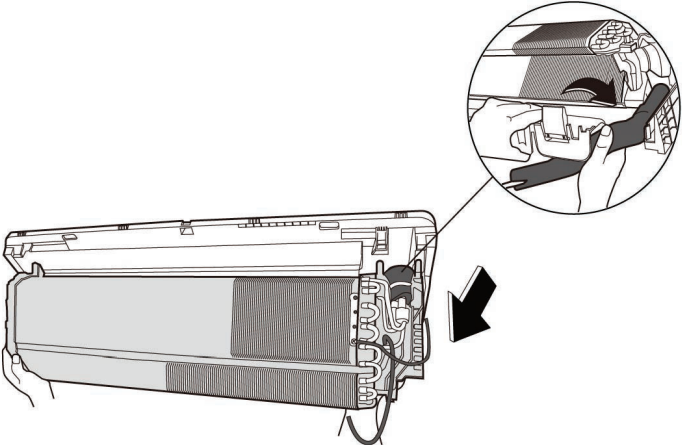
Evaporator

NOTE: Remove the front panel, electrical parts and the fan.

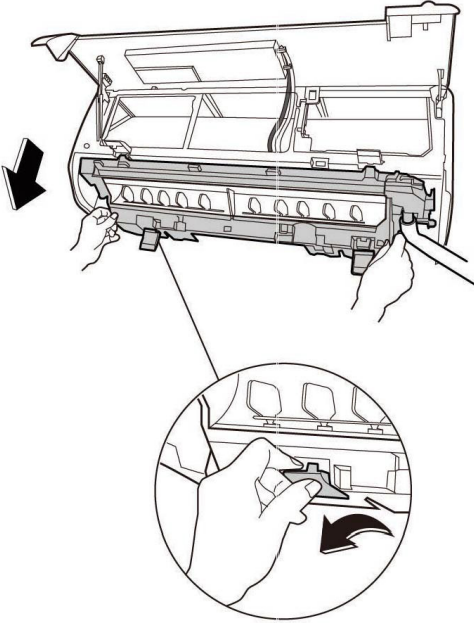
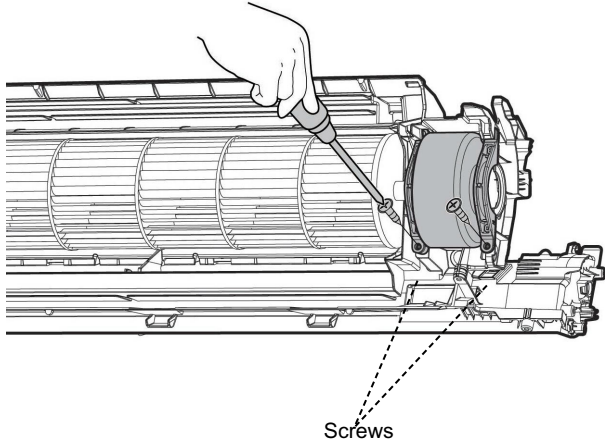
Procedure	Illustration
1) Remove the 2 screws and then the panel frame assembly.	
2) Disassemble the pipe clamp board.	

DISASSEMBLY INSTRUCTIONS (CONT)

Evaporator (CONT)

Procedure	Illustration
3) Remove the screw (1) on the evaporator located at the left fixed plate. 4) Remove the screw (1) on the evaporator located on the right side.	
5) Bend the piping carefully, separate the chassis assembly (above) and the evaporator, then remove the evaporator.	

DISASSEMBLY INSTRUCTIONS (CONT)**Fan Motor and Fan**

Procedure	Illustration
1) Open the two stop blocks of the chassis assembly (see adjacent picture). 2) Remove the chassis assembly (below) along the direction (see adjacent picture).	
3) Remove the two screws then remove the fan motor board.	

Manufacturer reserves the right to change, at any time, specifications and designs without notice and without obligations.

APPENDICES

Appendix 1

Table 13 — Temperature Sensor Resistance Value Table for T1, T2, T3, T4 (C--K)

°C	°F	K OHM	°C	°F	K OHM	°C	°F	K OHM	°C	°F	K OHM
-20	-4	115.266	20	68	12.6431	60	140	2.35774	100	212	0.62973
-19	-2	108.146	21	70	12.0561	61	142	2.27249	101	214	0.61148
-18	0	101.517	22	72	11.5	62	144	2.19073	102	216	0.59386
-17	1	96.3423	23	73	10.9731	63	145	2.11241	103	217	0.57683
-16	3	89.5865	24	75	10.4736	64	147	2.03732	104	219	0.56038
-15	5	84.219	25	77	10	65	149	1.96532	105	221	0.54448
-14	7	79.311	26	79	9.55074	66	151	1.89627	106	223	0.52912
-13	9	74.536	27	81	9.12445	67	153	1.83003	107	225	0.51426
-12	10	70.1698	28	82	8.71983	68	154	1.76647	108	226	0.49989
-11	12	66.0898	29	84	8.33566	69	156	1.70547	109	228	0.486
-10	14	62.2756	30	86	7.97078	70	158	1.64691	110	230	0.47256
-9	16	58.7079	31	88	7.62411	71	160	1.59068	111	232	0.45957
-8	18	56.3694	32	90	7.29464	72	162	1.53668	112	234	0.44699
-7	19	52.2438	33	91	6.98142	73	163	1.48481	113	235	0.43482
-6	21	49.3161	34	93	6.68355	74	165	1.43498	114	237	0.42304
-5	23	46.5725	35	95	6.40021	75	167	1.38703	115	239	0.41164
-4	25	44	36	97	6.13059	76	169	1.34105	116	241	0.4006
-3	27	41.5878	37	99	5.87359	77	171	1.29078	117	243	0.38991
-2	28	39.8239	38	100	5.62961	78	172	1.25423	118	244	0.37956
-1	30	37.1988	39	102	5.39689	79	174	1.2133	119	246	0.36954
0	32	35.2024	40	104	5.17519	80	176	1.17393	120	248	0.35982
1	34	33.3269	41	106	4.96392	81	178	1.13604	121	250	0.35042
2	36	31.5635	42	108	4.76253	82	180	1.09958	122	252	0.3413
3	37	29.9058	43	109	4.5705	83	181	1.06448	123	253	0.33246
4	39	28.3459	44	111	4.38736	84	183	1.03069	124	255	0.3239
5	41	26.8778	45	113	4.21263	85	185	0.99815	125	257	0.31559
6	43	25.4954	46	115	4.04589	86	187	0.96681	126	259	0.30754
7	45	24.1932	47	117	3.88673	87	189	0.93662	127	261	0.29974
8	46	22.5662	48	118	3.73476	88	190	0.90753	128	262	0.29216
9	48	21.8094	49	120	3.58962	89	192	0.8795	129	264	0.28482
10	50	20.7184	50	122	3.45097	90	194	0.85248	130	266	0.2777
11	52	19.6891	51	124	3.31847	91	196	0.82643	131	268	0.27078
12	54	18.7177	52	126	3.19183	92	198	0.80132	132	270	0.26408
13	55	17.8005	53	127	3.07075	93	199	0.77709	133	271	0.25757
14	57	16.9341	54	129	2.95896	94	201	0.75373	134	273	0.25125
15	59	16.1156	55	131	2.84421	95	203	0.73119	135	275	0.24512
16	61	15.3418	56	133	2.73823	96	205	0.70944	136	277	0.23916
17	63	14.6181	57	135	2.63682	97	207	0.68844	137	279	0.23338
18	64	13.918	58	136	2.53973	98	208	0.66818	138	280	0.22776
19	66	13.2631	59	138	2.44677	99	210	0.64862	139	282	0.22231

Appendix 2

Table 14 — Temperature Sensor Resistance Value Table for TP (T5)

° C	° F	K OHM	° C	° F	K OHM	° C	° F	K OHM	° C	° F	K OHM
-20	-4	542.7	20	68	68.66	60	140	13.59	100	212	3.702
-19	-2	511.9	21	70	65.62	61	142	13.11	101	214	3.595
-18	0	483	22	72	62.73	62	144	12.65	102	216	3.492
-17	1	455.9	23	73	59.98	63	145	12.21	103	217	3.392
-16	3	430.5	24	75	57.37	64	147	11.79	104	219	3.296
-15	5	406.7	25	77	54.89	65	149	11.38	105	221	3.203
-14	7	384.3	26	79	52.53	66	151	10.99	106	223	3.113
-13	9	363.3	27	81	50.28	67	153	10.61	107	225	3.025
-12	10	343.6	28	82	48.14	68	154	10.25	108	226	2.941
-11	12	325.1	29	84	46.11	69	156	9.902	109	228	2.86
-10	14	307.7	30	86	44.17	70	158	9.569	110	230	2.781
-9	16	291.3	31	88	42.33	71	160	9.248	111	232	2.704
-8	18	275.9	32	90	40.57	72	162	8.94	112	234	2.63
-7	19	261.4	33	91	38.89	73	163	8.643	113	235	2.559
-6	21	247.8	34	93	37.3	74	165	8.358	114	237	2.489
-5	23	234.9	35	95	35.78	75	167	8.084	115	239	2.422
-4	25	222.8	36	97	34.32	76	169	7.82	116	241	2.357
-3	27	211.4	37	99	32.94	77	171	7.566	117	243	2.294
-2	28	200.7	38	100	31.62	78	172	7.321	118	244	2.233
-1	30	190.5	39	102	30.36	79	174	7.086	119	246	2.174
0	32	180.9	40	104	29.15	80	176	6.859	120	248	2.117
1	34	171.9	41	106	28	81	178	6.641	121	250	2.061
2	36	163.3	42	108	26.9	82	180	6.43	122	252	2.007
3	37	155.2	43	109	25.86	83	181	6.228	123	253	1.955
4	39	147.6	44	111	24.85	84	183	6.033	124	255	1.905
5	41	140.4	45	113	23.89	85	185	5.844	125	257	1.856
6	43	133.5	46	115	22.89	86	187	5.663	126	259	1.808
7	45	127.1	47	117	22.1	87	189	5.488	127	261	1.762
8	46	121	48	118	21.26	88	190	5.32	128	262	1.717
9	48	115.2	49	120	20.46	89	192	5.157	129	264	1.674
10	50	109.8	50	122	19.69	90	194	5	130	266	1.632
11	52	104.6	51	124	18.96	91	196	4.849			
12	54	99.69	52	126	18.26	92	198	4.703			
13	55	95.05	53	127	17.58	93	199	4.562			
14	57	90.66	54	129	16.94	94	201	4.426			
15	59	86.49	55	131	16.32	95	203	4.294			
16	61	82.54	56	133	15.73	96	205	4.167			
17	63	78.79	57	135	15.16	97	207	4.045			
18	64	75.24	58	136	14.62	98	208	3.927			
19	66	71.86	59	138	14.09	99	210	3.812			

Appendix 3

Table 15 — Celsius to Fahrenheit

°C	°F	°C	°F	°C	°F	°C	°F	°C	°F
-5	23	21	69.8	51	123.8	82	179.6	113	235.4
-4	24.8	22	71.6	52	125.6	83	181.4	114	237.2
-3	26.6	23	73.4	53	127.4	84	183.2	115	239
-2	28.4	24	75.2	54	129.2	85	185	116	240.8
-1	30.2	25	77	55	131	86	186.8	117	242.6
0	32	25.5	77.9	56	132.8	87	188.6	118	244.4
0.5	32.9	26	78.8	57	134.6	88	190.4	119	246.2
1	33.8	27	80.6	58	136.4	89	192.2	120	248
1.5	34.7	28	82.4	59	138.2	90	194	121	249.8
2	35.6	29	84.2	60	140	91	195.8	122	251.6
2.5	36.5	30	86	61	141.8	92	197.6	123	253.4
3	37.4	31	87.8	62	143.6	93	199.4	124	255.2
3.5	38.3	32	89.6	63	145.4	94	201.2	125	257
4	39.2	33	91.4	64	147.2	95	203	126	258.8
4.5	40.1	34	93.2	65	149	96	204.8	127	260.6
5	41	35	95	66	150.8	97	206.6	128	262.4
6	42.8	36	96.8	67	152.6	98	208.4	129	264.2
7	44.6	37	98.6	68	154.4	99	210.2	130	266
8	46.4	38	100.4	69	156.2	100	212	131	267.8
9	48.2	39	102.2	70	158	101	213.8	132	269.6
10	50	40	104	71	159.8	102	215.6	133	271.4
11	51.8	41	105.8	72	161.6	103	217.4	134	273.2
12	53.6	42	107.6	73	163.4	104	219.2	135	275
13	55.4	43	109.4	74	165.2	105	221	136	276.8
14	57.2	44	111.2	75	167	106	222.8	137	278.6
15	59	45	113	76	168.8	107	224.6	138	280.4
16	60.8	46	114.8	77	170.6	108	226.4	139	282.2
17	62.6	47	116.6	78	172.4	109	228.2	140	284
18	64.4	48	118.4	79	174.2	110	230	141	285.8
19	66.2	49	120.2	80	176	111	231.8	142	287.6
20	68	50	122	81	177.8	112	233.6	143	289.4

